

Morphology-matched forward readout gates for Tau Core gravity: extending SPARC residual Papers 1–3 from inverse diagnostics to predeclared family tests

Jozsef Olcsak

2026

Abstract

Paper 1 found that residual-blind external structural disturbance labels are associated with increased low-acceleration residual scatter in SPARC galaxies. Paper 2 reversed the question and showed that fixed residual-shape features can infer those disturbance labels better than shuffled controls, while remaining non-unique with respect to MOND-simple and empirical RAR-like baselines. Paper 3 then built a candidate/control framework and introduced an endpoint-conditioned required- S_τ diagnostic: what preparation/readout factor would be needed to absorb a measured residual? This Paper 8 proposal turns that inverse diagnostic into a forward validation gate. The proposed gate predeclares a residual-blind observed morphology handle K_{obs} , audits whether it should be used directly or mapped to a readout-relevant proxy K_{readout} , maps K_{readout} to a Tau Core morphology-conditioned readout shell, computes a forward δg^K without using required S_τ as a predictor, and compares matched-family residual structure against wrong-family, shuffled-label, Newtonian baryonic, MOND-simple, and empirical RAR controls. The package also reaches a concrete NGC4013 milestone: a residual-blind source rule and formula-freeze gate now define an approximate mixed 4D readout protocol, MIXED_FORMULA_FREEZE_READY_NOT_RETROACTIVE_ENDPOINT, without converting the existing diagnostic score into retrospective validation. The permitted claim is methodological: the morphology-matched forward-readout gate is a falsifiable, Tau-Core-specific next endpoint for the Paper 1–3 program. This paper does not claim that Tau Core has been empirically validated, that a universal galaxy law has been derived, or that the matched-family endpoint has already beaten MOND/RAR on real SPARC data.

1 Purpose and claim boundary

The first three SPARC residual papers define a cautious progression. Paper 1 asks whether external structural disturbance is associated with residual scatter [5]. Paper 2 asks whether residual-shape features can recover external disturbance labels better than shuffled labels [6]. Paper 3 turns the association into a candidate/control framework and asks what S_τ would be required to absorb the observed residual [4].

That Paper 3 S_τ diagnostic is useful but partly inverse. It asks:

given the measured endpoint, what $S_\tau(R)$ would absorb it?

The present paper defines the next gate:

residual-blind observed morphology handle K_{obs}
→ readout-relevant proxy K_{readout}
→ forward readout δg^K before endpoint scoring.

The main claim is deliberately narrow:

The Tau Core gravity bridge can sharpen Papers 1–3 by replacing inverse required- S_τ residual absorption with a morphology-matched forward-readout candidate/control gate.

This paper does not claim:

- that Tau Core is proven;
- that the morphology-matched gate has already won on real SPARC endpoints;
- that MOND, RAR, or Newtonian baryonic baselines have been superseded;
- that non-axisymmetric morphology classes can be fully tested from one-dimensional rotation curves alone;
- that the morphology families below are the final physical taxonomy.

2 From Paper 3 inverse diagnostics to forward readout

Paper 3 used an endpoint-conditioned diagnostic of the form

$$S_{\tau,\text{req}}(R) = \frac{V_{\text{obs}}(R)/V_N(R) - 1}{\alpha \ln(1 + a_0/a_N(R))}.$$

This quantity is descriptive. Because it uses V_{obs} , it cannot be treated as an independent predictor. Its correct role is to identify what kind of future nonleaky rule would be required.

The forward replacement is:

$$K_{\text{obs}} \longrightarrow K_{\text{readout}} \longrightarrow \mathcal{F}_{K_{\text{readout}}} \longrightarrow \delta g^{K_{\text{readout}}}(R) \longrightarrow \text{Score}(g_{\text{Newt}} + \delta g^{K_{\text{readout}}}).$$

Here K_{obs} is the residual-blind observed 4D morphology handle, K_{readout} is the predeclared readout-relevant morphology proxy, $\mathcal{F}_{K_{\text{readout}}}$ is the selected readout family, and $\delta g^{K_{\text{readout}}}$ is computed without using $S_{\tau,\text{req}}$, endpoint residual gain, or post-hoc family selection. The equality $K_{\text{obs}} = K_{\text{readout}}$ is allowed but not assumed; this is a Tau Core bridge consequence of treating morphology as a projection/readout structure rather than only a present-day visual class.

3 MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

For a galaxy g , let

$$K_{\text{obs},g}$$

be its residual-blind observed 4D morphology handle, and let

$$K_{\text{readout},g} \in \mathcal{K}$$

be the predeclared readout-relevant morphology proxy used for formula-shell selection. Let $\mathcal{F}_K$ denote the Tau Core readout shell assigned to readout proxy class K . Let

$$R_{g,K}$$

be the residual score after applying the forward readout family $\mathcal{F}_K$ to galaxy g under a frozen geometry and amplitude policy.

The matched-family score is

$$R_{g,\text{matched}} := R_{g,K_{\text{readout},g}}.$$

The wrong-family score is

$$R_{g,\text{wrong}} := \frac{1}{|\mathcal{K}| - 1} \sum_{K \neq K_{\text{readout},g}} R_{g,K}.$$

The central paper-level endpoint can be written as

$$\Delta_{\text{matched}} = \text{RMS}_g(R_{g,\text{wrong}}) - \text{RMS}_g(R_{g,\text{matched}}).$$

An equivalent rank endpoint is

$$\text{rank}_g(K_{\text{readout},g}) = \text{rank}(R_{g,K_{\text{readout},g}} \text{ among } \{R_{g,K} : K \in \mathcal{K}\}).$$

The gate passes only if the predeclared matched family improves residual structure more than:

1. wrong morphology families;
2. shuffled morphology labels;
3. Newtonian baryonic baseline;
4. MOND-simple baseline;
5. empirical RAR-like baseline.

This is Tau-Core-specific in a way that ordinary MOND/RAR comparisons are not: the discriminating claim is not merely lower residuals, but morphology-indexed readout specificity.

MORPHOLOGY-MATCHED-FORWARD-READOUT-GATE

- 
1. residual-blind morphology label
 2. formula-shell selection
 3. geometry and amplitude discipline
 4. matched-vs-wrong family endpoint
 5. shuffled-K null
 6. baseline comparison

Figure 1: The proposed morphology-matched forward-readout gate. The labels and family choices must be fixed before endpoint scoring.

4 Morphology-family registry

The current gravity bridge centralizes a morphology-conditioned TGP-like readout lane. The corresponding first-pass family registry is not a universal galaxy law. It is a set of candidate readout shells whose selection must be controlled by residual-blind morphology evidence. In this paper, K_{obs} denotes the apparent 4D morphology handle and K_{readout} denotes the readout-relevant proxy used to select the shell. Thin/thick disk, bar, ring, compact, tail, and lopsided labels are therefore treated as projected 4D morphology handles unless a source-review or bridge argument promotes them to readout-relevant proxies.

Table 1: Claim-bounded morphology-family registry for the forward-readout gate.

Family	Morphology label	Formula shell	First-pass status
K_{tail}	Scale-tail spiral / diffuse LSB disk	$\sigma_{\text{morph}}(R) \sim A_K/(R + R_c)^2$	1D testable
K_{exp}	Regular exponential disk	$\sigma_{\text{morph}}(R) \sim A_K e^{-R/R_d}$	1D testable
K_{compact}	Compact finite support	$I_{\text{morph}}(R) \rightarrow I_\infty$ and $\delta g_R \sim -I_\infty/R^2$	1D testable
K_{ring}	Ring or resonance feature	localized $\sigma_{\text{morph}}(R)$ near R_{ring}	1D proxy
K_{flare}	Thick or flared disk	projected thickness/flaring kernel	1D proxy
$K_{m=2}$	Barred spiral / $m = 2$ mode	harmonic $\sigma_{\text{morph}}(R, \phi)$	velocity-field preferred
$K_{m=1}$	Lopsided / $m = 1$ mode	harmonic $\sigma_{\text{morph}}(R, \phi)$	velocity-field preferred

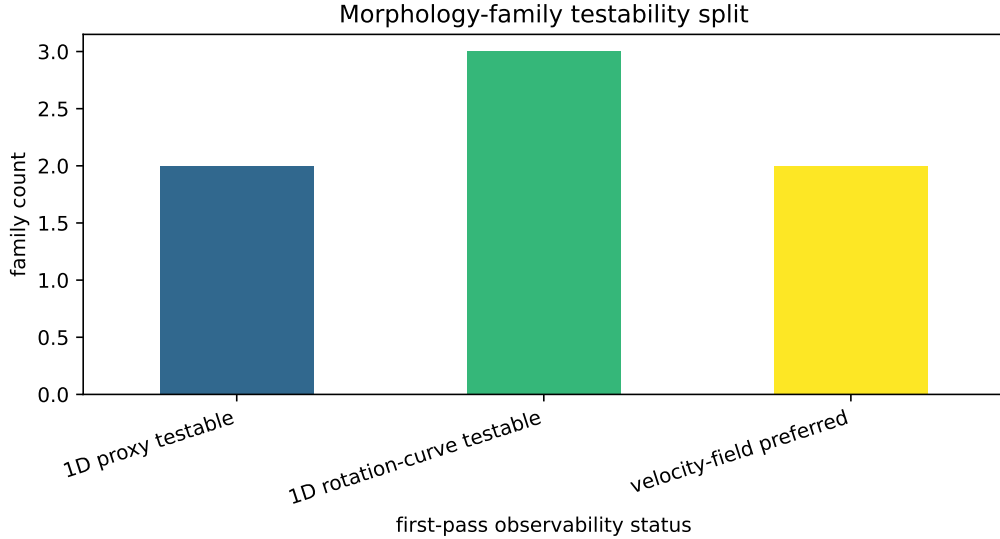


Figure 2: First-pass observability split for the current family registry. SPARC one-dimensional rotation curves are most suitable for axisymmetric or radially localized families; barred and lopsided branches require velocity-field follow-up for a full test.

5 Candidate/control continuity from Paper 3

The gate preserves the Paper 3 candidate/control discipline. It does not discard difficult controls after seeing endpoints. The same object may carry a different role under the forward gate:

candidate/control role $\rightarrow$ predeclared morphology row $\rightarrow$ matched/wrong/shuffled score.

Table 2: Paper 3 roles carried into the Paper 8 forward-gate protocol.

Galaxy	Paper 3 role	Paper 8 role
DDO126	positive anchor	matched-family candidate after residual-blind morphology labeling
DDO50	quiet control	false-positive suppression control
WLM	geometry stress	observability-caveated endpoint row
DDO168	ladder support	candidate/control consistency row
DDO154	countercontrol	wrong-family and rank-sensitivity stress row

The positive paper-level endpoint is not:

Tau Core fits every galaxy better.

The first meaningful endpoint is:

the correct predeclared morphology family ranks better than wrong families.

This is why shuffled- K nulls are mandatory. Without the shuffled-label null, a morphology-indexed family could accidentally behave like a generic flexible residual smoother.

6 Synthetic protocol fixture

This package includes a synthetic score fixture to verify the endpoint logic. It is not real SPARC evidence. It exists only so the public tests can check that the manuscript, tables, and figure pipeline agree about the intended endpoint.

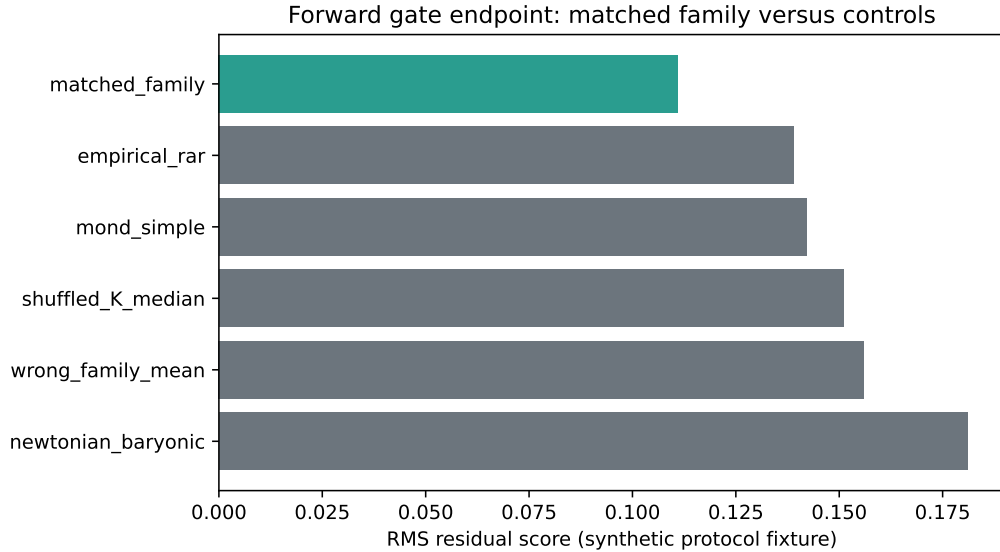


Figure 3: Synthetic protocol fixture for the endpoint schema. A real pass would require the matched-family score to beat wrong-family and shuffled-label controls, then be compared to Newtonian baryonic, MOND-simple, and empirical RAR baselines.

7 Available-data Tau-proxy preflight

The repository also includes a first available-data preflight of the matched-family idea. This is not the final Paper 8 endpoint, because the family functions are still proxy functions built from

the locally available Tau-proxy channel rather than the final morphology-specific δg^K readout shells. The preflight nevertheless tests the important next question: whether a metadata-proxy morphology family assigned before scoring performs better than wrong Tau-proxy families on held-out galaxies.

In the current 175-galaxy point-level packet, family amplitudes are fit on the train split only. On the 44-galaxy holdout split, the morphology-matched proxy beats the wrong-family mean in 0.568 of galaxies, beats the TPG/v6 comparator in 0.636 of galaxies, and beats the MOND comparator in 0.614 of galaxies. A deterministic 1000-permutation shuffled-label null gives a holdout null mean of 0.0133 for the mean matched-minus-wrong endpoint, compared with the observed value -0.0326 , with $p \simeq 0.264$ for shuffled labels being at least as good. Thus the available proxy signal is positive but not yet a discovery-level morphology-specific result. These numbers are recorded as a preparation result, not as empirical validation of Tau Core. They motivate replacing the generic Tau proxy with source-native morphology-family readout shells and a stronger shuffled-label null in the final endpoint.

A second preflight replaces the single available Tau-proxy channel with different radial formula-shell proxies assigned by morphology class. This is closer in form to the intended Paper 8 family test, but the current available-data version is weaker: on the same 44-galaxy holdout split it beats the wrong-shell mean in 0.500 of galaxies, beats TPG/v6 in 0.477 of galaxies, and has a shuffled-null $p \simeq 0.479$ for the mean matched-minus-wrong endpoint. This negative preflight result is useful. It indicates that merely assigning naive radial shell shapes is not sufficient; the final endpoint needs source-native morphology labels, dimensionally audited 4D readout shells, and a nonleaky amplitude policy.

The source-native bridge formula preflight then consumes a declared morphology-parameter manifest and evaluates the concrete morphology formulas as δv_K^2 readout kernels: the scale-tail $n = 2$ branch, the finite exponential disk Freeman/Bessel kernel, the compact finite-source exterior response, and the thick/flared damped vertical-kernel response. With the same available-data morphology labels and train-only amplitudes, the 44-galaxy holdout split gives a much stronger morphology-specific endpoint: the matched bridge formula beats the wrong-formula mean in 0.886 of galaxies, with a 1000-permutation shuffled-label null giving $p \simeq 0.002$ for the mean matched-minus-wrong endpoint. However, it beats TPG/v6 and MOND in only 0.477 of holdout galaxies each, and its mean matched-minus-baseline residual is still worse than both baselines. Thus the bridge-formula result supports the Paper 8 morphology-family specificity hypothesis, but it is not an empirical claim that Tau Core outperforms TPG/MOND.

A readout-mixture proxy diagnostic then tests the natural bridge refinement that a galaxy may carry a mixture of readout components rather than a single hard family. This diagnostic keeps the same concrete bridge kernels and uses residual-blind available-data proxy weights. The current holdout result is negative: the mixture beats the hard matched family in only 0.455 of galaxies, beats TPG/v6 in 0.500, and beats MOND in 0.455. This is a useful failure map, not a defeat of the bridge idea. It shows that a mixture layer is only meaningful if its weights are supplied by accepted morphology-memory/readout observables, not by coarse present-day 4D proxy heuristics.

The package therefore adds a readout-state vector intake gate. This gate does not score rotations. It lists the residual-blind source observables required before the mixture weights can be promoted into an accepted Tau-side readout-state vector: disk scale/decomposition, outer-disk or HI tail support, compact-core support, vertical flare/projection evidence, morphology-memory/history evidence, and a source-normalization rule. Under the strict accepted-input rule, the current package remains blocked: the memory/history layer is proxy-only and the Tau-side normalization rule is now present only as a theory-conditional candidate, not as an accepted physical normalization law.

A confidence diagnostic on the manifest shows why the baseline claim remains blocked. Restricting the holdout set to confidence ≥ 0.75 preserves the matched-vs-wrong signal, with matched formulas beating wrong formulas in 0.906 of galaxies, but the TPG/MOND comparison is still mixed. Removing low-inclination or large-distance-error objects improves some baseline comparisons, so data quality matters. It is not the whole bottleneck: the remaining blockers are source-native morphology-scale extraction, amplitude policy, and normalization of the bridge formula kernels.

A baseline-success morphology audit then asks the complementary control question: where TPG/MOND/Newton/RAR already work, what Tau Core morphology/readout regime might that indicate? In the current holdout split, TPG/v6 wins are concentrated in scale-tail/irregular rows and have zero current-proxy/readout-memory agreement, suggesting a possible smooth closure-like or memory-integrated readout regime. Tau-matched wins have higher current-readout consistency, while MOND wins are compatible with simpler radial low-acceleration or diffuse-disk scaling regimes. Where Newtonian wins in the smaller conventional tables, the claim-safe Tau Core interpretation is a quiet or regular baryonic-readout regime, not a refutation of morphology specificity elsewhere. Thus baseline success is not discarded. Newtonian-good rows become quiet baryonic-readout controls; MOND/RAR-good rows become scalarized radial-readout controls; TPG-good rows become closure-like or memory-integrated readout controls; and Tau-family-good rows become cases where the present morphology proxy may be closer to K_{readout} . This makes the final question sharper: whether baseline success zones and Tau-family success zones map to predeclared readout regimes rather than to post-hoc model choice.

The same audit gives a conditional limiting-regime interpretation of the comparison baselines. In a quiet, current-regular baryonic-readout state with small α_τ and suppressed quotient-visible morphology residual, the Tau Core readout should reduce to the Newtonian weak-field limit, $g_{\text{eff}} \rightarrow g_{\text{Newt}}$. In a scalarized radial low-acceleration state, where morphology dependence is weak or averaged and the needed scale is supplied by Tau-side closure/readout data rather than inserted as a fit law, a MOND/RAR-like solved response can be a special effective limit. In a smooth closure-like or memory-integrated state with an outer logarithmic or near-flat-speed solved branch, a TPG-like formula can be the natural effective readout. If the same residual descends through a metric-compatible and gauge-safe weak-field readout map, the branch is RMOND-facing, meaning suitable for relativistic/metric audit, not yet a derived relativistic MOND theory. This is the intended bridge statement: successful Newtonian, MOND/RAR, TPG, and RMOND-facing behavior should be explained as conditional Tau Core readout limits when their regime assumptions hold, while the endpoint still tests whether these regimes were predeclared rather than selected after seeing residuals.

The bridge status of these limiting forms is deliberately uneven. The Newtonian limit is formula-derived when $\delta g_{\text{morph}} \rightarrow 0$. The TPG-like shape is formula-derived under the $n = 2$ source-tail gate,

$$\sigma_{\text{morph}} \sim \frac{A}{r^2} \implies \delta g \sim \frac{A}{r} \implies \delta \Phi \sim A \log r.$$

The fixed historical TPG branch remains formula-conditional because the finite-load constants, branch choice, and historical normalization require additional Tau-side closure/readout input. The MOND/RAR-like scaling is also formula-conditional:

$$A^2 = GM_b g_* \implies \delta g = \sqrt{g_N g_*}.$$

Here $g_* := \lambda_* \ell_*$ is an internal Tau-side acceleration-scale object only if λ_* is a residual-blind closure/readout stiffness with dimension T^{-2} and ℓ_* is an admissible readout length. This is not a numerical a_0 validation, not a universal MOND-scale derivation, and not an empirical baseline-

superiority claim. The RMOND-facing branch remains readout-form-only until a covariant field equation, conservation audit, and lensing sector are supplied.

This gives a conditional baseline-selection theorem. Let R_g be a residual-blind readout-regime label assigned to a galaxy before rotation endpoint scoring. If R_g satisfies the quiet baryonic-readout assumptions, then the suppressed-residual limit gives the Newtonian branch. If R_g satisfies the scalarized radial low-acceleration assumptions and an internal $g_* = \lambda_* \ell_*$ scale has been supplied, then the solved response has a MOND/RAR-like branch. If R_g satisfies the smooth closure/readout source-tail assumptions, in particular the $n = 2$ tail $\sigma_{\text{morph}} \sim A/r^2$, then the solved response has a TPG-like logarithmic or near-flat-speed branch. If R_g satisfies the gauge-safe metric-descent assumptions, then the branch is RMOND-facing as a weak-field readout form. The proof is by substitution of the corresponding regime assumptions into the bridge formula-status ladder above. The theorem does not choose a baseline after seeing residuals: if several regime predicates hold, the galaxy is an overlap/control case; if none holds, the current bridge has not explained that baseline success.

The resulting empirical target is a morphology-information-gain test rather than a full high-dimensional morphology fit. A complete galaxy morphology would require many source variables, including disk scale, compact support, bar strength, ring structure, thickness or flare, HI tails, projection, environment, and morphology-memory indicators. The claim-safe test is lower-dimensional: as residual-blind morphology/readout information is refined from a coarse K_{obs} label, to source-reviewed K_{readout} , to a readout-state vector, to source-native scales and amplitudes, the forward Tau Core readout should improve in predeclared scores. A pass is not required to be monotonic for every galaxy, but the population-level trend should improve matched-vs-wrong rank, shuffled-label separation, residual RMS, baseline competitiveness, or readout-regime classification. A fail occurs if additional morphology information gives no improvement, or if improvement appears only when rotation residuals are used to choose labels, weights, scales, or gates.

The repository now includes a first executable version of this test. It assembles four currently available levels: $L0$ coarse formula-shell proxy, $L1$ source-native bridge formula shell, $L2$ readout-state mixture proxy, and $L3$ source-native shells with train-selected normalization. On the current holdout split, the $L0 \rightarrow L1$ transition strengthens family specificity but does not improve raw RMSE; $L1 \rightarrow L2$ is mixed or negative because the mixture weights are still proxy-only; and $L2 \rightarrow L3$ improves the proxy mixture layer. The associated full-sample source expansion records 175 SPARC rows, 75 S4G scale-radius candidates, 31 DustPedia source-candidate matches, 171 SPARC HI-ready galaxies, two PHANGS public-sample matches, 172 $L2$ tail candidates, 48 $L2$ compact candidates, 19 $L2$ bar candidates, zero MUSE-ready velocity-field matches, and no endpoint-ready accepted readout-state vector components. A separate $L2$ weight-intake preflight converts these source candidates into residual-blind candidate readout weights for 174 of 175 galaxies, with one uninformative fallback; nonzero candidate support appears for 172 tail, 75 exponential-disk, 48 compact, and 107 thick/flared channels. These weights are not accepted Tau-side readout-state weights and are not endpoint scores. When scored as an endpoint stress test, the $L2$ intake layer remains mixed/negative on holdout: it beats the old $L2$ mixture proxy in 0.409 of galaxies, beats the hard source-native family in 0.409, beats TPG/v6 in 0.477, beats MOND in 0.432, and has median intake-minus-old- $L2$ RMSE +0.847.

The package then introduces a theory-conditional residual-blind Tau-side source-normalization candidate,

$$\hat{K}_{gK}(r) = \frac{K_{gK}(r)}{\text{median}_r |K_{gK}(r)|}, \quad c_g = \frac{\text{median}_r \max(v_{v6}^2 - v_n^2, 0)}{\text{median}_r v_{v6}^2},$$

$$\delta v_{gK}^2(r) = \sigma_K e_{gK} w_{gK} c_g \text{median}_r (v_n^2) \hat{K}_{gK}(r).$$

Here w_{gK} is the residual-blind L2 intake weight, e_{gK} is the residual-blind source-evidence gate, and σ_K is a predeclared readout-orientation sign: compact and tail components are positive, while exponential-disk and thick/flared components are negative. A separate derivation manifest records these signs and gates before endpoint scoring and audits the rule as dimensionally consistent for a δv^2 candidate; the manifest marks the orientation signs as theory-conditional bridge derivations and the proxy attenuation as the coarse executable representative of the conservative Tau-side readout-admission product, with standard proxy value $0.70 \times 0.70 \times 0.85 \times 0.85 = 0.354025$. No observed velocity endpoint, rotation residual, required- S_τ diagnostic, or best-family choice enters this normalization. On holdout, this source-normalized L2 preflight beats the old L2 intake endpoint in 0.568 of galaxies, beats TPG/v6 in 0.455, beats MOND in 0.545, and has median source-normalized-minus-old-L2 RMSE -0.272 . This is an improvement over the raw L2 intake endpoint, but not an accepted normalization law. A post- E_τ freeze-readiness audit still blocks endpoint use for all 175 galaxies, but the blocker has changed: the proxy-gate blocker is resolved for all 175 rows by the derived coarse-grid E_τ product, missing Tau-side normalization blocks 0 rows, and a source-native orientation-promotion gate now promotes three of four family orientations. In galaxy-level active-component terms, 67 of 175 rows are orientation-ready and 108 remain orientation-blocked. A stricter memory/projection gate then splits those 67 rows into 47 projection-blocked rows, 19 memory/history-blocked rows, and one memory/projection-ready candidate that still lacks accepted per-galaxy q_i and normalization-law acceptance. Dominant components have source-candidate support in 141 rows, are proxy/partial-only in 31 rows, and lack dominant source support in three rows. The preserved boundary is therefore sharper: source coverage plus a residual-blind normalization candidate is not yet enough; the remaining thick/flared orientation rows, projection caveats, memory/history acceptance, accepted per-galaxy evidence assignments, and an accepted normalization law are still required before endpoint freeze. This is a failure map and data-acquisition guide, not a monotonicity proof.

The repository also records a lane-expansion audit so that the sample can grow without weakening the accepted-lane claim. Under the current residual-blind gates, only one galaxy is a strict memory/projection-ready candidate, but 66 additional galaxies are caution/proxy-supported because they are orientation-ready while still carrying projection or memory/history caveats. Thus 67 of 175 galaxies are analysis-includable in strict-plus-caution support lanes, while 108 remain acquisition-required. The caution lane is explicitly not accepted evidence; it is a support/sensitivity lane for deciding which projection, vertical-structure, and memory/history source data should be acquired next.

When the existing endpoint diagnostics are sliced by these lanes, the current holdout strict-plus-caution subset contains 16 galaxies. On that subset, the source-native hard-family formula keeps a morphology-specific signal, beating the wrong-family mean in 0.8125 of galaxies, but it is only mixed against TPG/v6 and MOND, with win fractions 0.5000 and 0.4375. The Tau-side evidence-measure L2 normalization is weaker on the same lane, beating TPG/v6 in 0.3750 and MOND in 0.3125 of galaxies. This is not a validation failure and not a discovery claim; it is a preparation diagnostic showing that the expanded lane is useful for sensitivity and acquisition planning while baseline-superiority remains blocked.

Splitting the caution lane by allowed use sharpens the failure map. In the holdout projection-caveat sub-lane, the source-native hard-family formula still beats the wrong-family mean in 0.7857 of galaxies, while the Tau-side evidence-measure L2 normalization beats TPG/v6 in only 0.3571 and MOND in only 0.2857. Thus the present weakness is not primarily the disappearance of morphology-family specificity; it is the projection/scale and source-normalization layer needed to turn that specificity into a competitive solved response.

Two repair audits make this more operational. The projection/scale repair map finds 71 of 175

rows with no projection/scale repair required, 34 requiring vertical-geometry source evidence, 26 requiring inclination/projection review, 30 requiring new distance/scale source support, and 14 that are plausibly repairable with existing scale-source information plus a distance/scale audit. A source-normalization failure-mode audit then shows that, inside the holdout projection-caveat sub-lane, 7 of 14 rows are projection-scale normalization failures: the hard-family morphology-specific signal is present, but the Tau evidence L2 normalization fails to transfer it into a baseline-competitive solved response. This identifies a concrete repair target without changing any endpoint gate.

The bridge-level promotion target is not to defend a universal proxy constant. It is to replace the coarse proxy bin by a residual-blind evidence measure $e_{gK} = E_\tau(g, K)$ built from source presence, geometry relevance, projection safety, morphology-memory reliability, and resolution adequacy. In the current Tau Core bridge derivation, accepted/proxy/missing are treated as coarse readout-admission states, proxy is the minimal admitted nonmissing state, and conservative three-status mass normalization gives $E_{\text{proxy}}^* = 1/3$. Combined with the standard proxy template this gives $\epsilon_* = 0.157399\dots$, whose coarse-grid representatives are strong proxy 0.85 and ordinary proxy 0.70. The resulting standard proxy product is 0.354025. This is stronger than a hand-fixed 0.35 protocol constant, while still not a claim that the conservative three-status evidence geometry is the unique final Tau-side evidence law.

The repository now includes a first executable version of this promotion target. For all nonzero L2 components, it builds

$$E_\tau = q_{\text{source}} q_{\text{geometry}} q_{\text{projection}} q_{\text{memory}} q_{\text{resolution}}.$$

On the full sample, this audits 406 components, including 105 proxy/partial components. The median proxy value is $E_\tau = 0.354025$, with median proxy-minus-fixed-0.35 equal to +0.004025. Replacing the fixed proxy-bin gate by this $E_\tau(g, K)$ candidate leaves the holdout win fractions unchanged relative to the fixed-gate source-normalized preflight: it beats the old L2 intake endpoint in 0.568 of galaxies, TPG/v6 in 0.455, and MOND in 0.545, while the median E_τ -minus-old-L2 RMSE is -0.236 . This is not model selection. It shows that the fixed proxy gate can be promoted to a residual-blind evidence-measure candidate without using endpoint residuals. The proxy ladder is now derived inside the current conservative readout-admission geometry; the remaining caveat is that the galaxy/component q_i assignments are still theory-candidate source/readout assignments rather than accepted source-native observables.

The executable audit also records per-factor derivation status. Accepted and missing source gates are definition-derived limit cases of the evidence measure. Proxy/partial rows are kept in a separate theory-candidate factor-geometry status until their source, geometry, projection, memory, and resolution factors are fixed by residual-blind Tau-side readout rules rather than endpoint performance.

A sensitivity audit varies only the conditional orientation signs and source-evidence gates. It is not a model-selection step. On holdout, the primary proxy-gate value $e_{\text{proxy}} = 0.35$ beats the old L2 intake endpoint in 0.568 of galaxies. Removing proxy support gives 0.545. Raising proxy support to unity gives 0.523 against the old L2 intake endpoint and 0.523 against TPG/v6, but this stronger TPG-facing number cannot be used as a selected endpoint because the gate is fixed by the conservative readout-admission derivation rather than by endpoint performance. All-positive orientation signs reduce the old-L2 win fraction to 0.477, and all-negative signs reduce it to 0.432. Thus the orientation structure is not cosmetic; after the proxy-gate derivation, the source-native orientation gate partially promotes the signs, and the remaining freeze blockers are the thick/flared orientation source layer plus accepted per-galaxy evidence and morphology-memory/projection assignments.

An amplitude-policy diagnostic supports the same conclusion. Keeping the source-native bridge kernels fixed, the family-unconstrained policy gives the strongest matched-vs-wrong signal but remains weak against TPG/MOND baselines. A simple family-to-global shrinkage policy preserves most of the matched-vs-wrong signal, 0.864 on holdout, while raising the MOND win fraction to 0.636 and reducing the mean MOND gap. This is not a selected endpoint claim; it identifies amplitude normalization as a required Tau-side modeling gate before any baseline-superiority claim is allowed.

A shrinkage-path scan makes this more precise. Varying the family amplitude weight from 0 to 1 shows that the current best compromise lies near family weight 0.4–0.5. At weight 0.5, the holdout matched-vs-wrong fraction is 0.864 and the MOND win fraction is 0.636, while TPG/v6 remains at 0.477. This identifies the normalization target for the next Tau-side gate: a physical source-normalization rule should recover a partially pooled family amplitude, not a fully free family amplitude chosen only for endpoint fit.

A train-selected shrinkage diagnostic reduces the post-hoc risk in that conclusion. Four selection rules using train split metrics only all select family weight 0.40. Evaluated on the 44-galaxy holdout split, this train-selected policy keeps matched-vs-wrong specificity at 0.818, beats MOND in 0.614 of galaxies, and remains mixed against TPG/v6 at 0.477. This supports a stable partially pooled normalization range, but it is still not a validated amplitude-normalization law or a baseline-superiority endpoint.

A family-breakdown diagnostic then asks where this signal lives. Under the train-selected 0.40 shrinkage policy, the holdout matched-vs-wrong specificity is strongest for the scale-tail spiral and thick/flared families, both at 0.900. The thick/flared row is also competitive against both TPG/v6 and MOND in this diagnostic, while the scale-tail row remains MOND-competitive but TPG-blocked. The exponential-disk and compact finite-source rows are weaker under the current available-data manifest. This is useful as a failure map: it points to family-specific source normalization and morphology-observable extraction as the next modeling gates, not to a validated family-level empirical claim.

An observable-quality diagnostic separates part of that failure map. Joining the train-selected scores to the residual-blind morphology manifest shows that, on holdout, the thick/flared family is the cleanest current-best-case row, while the compact finite-source, exponential-disk, and scale-tail rows are quality-limited under the available-data proxies. In the clean holdout manifest subset, matched formulas beat wrong formulas in 0.900 of galaxies and beat MOND in 0.750, whereas quality-caveated rows fall to 0.750 and 0.500. This does not rescue the endpoint by filtering after the fact; it identifies residual-blind morphology-observable extraction as a required predeclared gate for the next run.

A predeclared quality-gate diagnostic makes that next run more concrete. Under the fixed train-selected shrinkage policy, the clean-manifest gate and the confidence- ≥ 0.75 -and-clean gate each retain 20 holdout galaxies across all four formula families, with matched-vs-wrong specificity 0.900, TPG/v6 win fraction 0.500, and MOND win fraction 0.750. A broader no-large-distance-error gate retains 26 holdout galaxies and remains candidate-level, with matched-vs-wrong 0.885, TPG/v6 0.538, and MOND 0.731. These are not selected discovery endpoints; they are candidate observability rules that must be declared before endpoint scoring in a future Paper 8 run.

The corresponding shuffled-label null reveals the tradeoff. On holdout, the full all-galaxy gate has the strongest shuffled-null support for morphology specificity, with $p \simeq 0.036$ for the beats-wrong fraction and $p \simeq 0.013$ for the mean matched-minus-wrong endpoint. The no-low-inclination gate gives similar values, $p \simeq 0.041$ and $p \simeq 0.011$. Cleaner gates improve the MOND/TPG comparison but have weaker shuffled-null support because the sample shrinks. Thus Paper 8 should not silently choose the cleanest-looking subset; it should predeclare whether the endpoint prioritizes

observability-clean baseline competitiveness or larger-sample morphology-label specificity.

Combining these diagnostics gives a concrete preparation-state endpoint decision matrix. The current natural primary endpoint candidate is the no-low-inclination gate: it retains 35 holdout galaxies, remains baseline-competitive, and has strong shuffled-null support. The full all-galaxy gate should remain a fuller-sample morphology-specificity support lane, while the no-large-distance-error gate is the strongest secondary baseline-competitiveness lane. These roles must be predeclared before any future endpoint scoring; they are not retrospective proof of Tau Core.

The repository therefore records a predeclared endpoint protocol sheet. It freezes the current preparation-state choices: primary lane no-low-inclination, fuller-sample support lane all, secondary baseline lane no-large-distance-error, train-selected family-to-global shrinkage with weight 0.40, residual-blind morphology-family assignment, matched-vs-wrong plus shuffled-family null specificity metrics, and TPG/v6 plus MOND comparator metrics. It also records forbidden inputs, including endpoint residual gain, required- S_τ , post-hoc gate choice, and per-galaxy amplitude tuning. This is the guardrail for the next run, not a claim that the present diagnostic run validates Tau Core.

8 Minimal numerical protocol

The minimal public endpoint protocol is:

1. Assign an external or residual-blind observed 4D morphology handle $K_{\text{obs},g}$ to each galaxy.
2. Predeclare whether $K_{\text{obs},g}$ is used directly or mapped to a readout-relevant proxy $K_{\text{readout},g}$ using only residual-blind source evidence.
3. Select a formula shell $\mathcal{F}_{K_{\text{readout},g}}$ from the predeclared registry.
4. Fix geometry parameters from morphology, not from residual endpoints.
5. Use either a frozen amplitude rule or a tightly bounded amplitude fit.
6. Compute the matched-family score.
7. Compute wrong-family scores for all other candidate families.
8. Run shuffled- K nulls.
9. Compare against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

The information-gain version of the protocol repeats this endpoint at predeclared morphology-information levels:

- $L0 : K_{\text{obs}},$
- $L1 : K_{\text{readout}},$
- $L2 : \mathbf{q}_{\text{readout}},$
- $L3 : \text{source-native scales and normalization},$
- $L4 : \text{velocity-field / HI / history-enriched data}.$

The target is not a per-galaxy free morphology fit. The target is a population-level improvement curve under residual-blind information upgrades.

For SPARC-only first-pass work, the cleanest initial family set is:

$$\{K_{\text{tail}}, K_{\text{exp}}, K_{\text{compact}}, K_{\text{ring}}, K_{\text{flare}}\}.$$

The $m = 1$ and $m = 2$ branches should be treated as caveated proxy rows in one-dimensional rotation curves and as real tests only when velocity-field products are available.

9 Why this would be a stronger Tau Core test

The new gate is stronger than a residual association because it creates a family-selection burden. A theory with morphology-indexed readout families should not only reduce residuals; it should reduce them more when the correct morphology family is used than when a wrong family is used. MOND-simple and empirical RAR baselines are strong comparators, but they do not naturally provide a predeclared morphology-family rank test of this kind.

The result that would matter is therefore:

$$\Delta_{\text{matched}} > 0$$

with a rank distribution in which $K_{\text{readout},g}$ is systematically near the top, under residual-blind observed morphology handles, predeclared readout-proxy rules, and shuffled-label controls. If this fails, the bridge loses specificity. If it passes, the Paper 1–3 residual-disturbance program becomes a forward-predictive morphology-readout program.

The current mixed-readout replay/holdout check is a useful negative-control example of this burden. Using only the fresh replay/holdout rows, NGC5907 with its source-frozen projection mixed formula and NGC7331 with the V2 fractional-onset replay manifest both improve over Newtonian, TPG/v6, MOND, and the exponential-disk carrier. However, the matched labels do not beat the wrong-label controls: the matched permutation ranks second of two and the best shuffled assignment is slightly better by about 0.044 km s^{-1} in mean RMSE. This is preserved as a small- N negative specificity result. It does not refute the bridge, but it shows that the current two mixed attenuation formulas are not yet readout-specific enough to support a replay/holdout specificity claim.

The package therefore adds a residual-blind source-side kernel-observable separation gate, with status `SOURCE\_KERNEL\_OBSERVABLE\_SEPARATION\_PASS`. This gate does not read observed velocities, residuals, RMSE ranks, or required- S_τ diagnostics. It compares only source-native dimensionless features. On the same two fresh rows, the source fingerprints separate cleanly: NGC5907 is closer to the projection mixed prototype than to the vertical/outer-warp V2 prototype (0.921 versus 0.419), while NGC7331 is closer to the vertical/outer-warp V2 prototype than to the projection prototype (0.978 versus 0.177). The interpretation is therefore sharper than “the source labels are too ambiguous.” At this stage the source observables do separate the lanes; the bottleneck is the source-to-kernel map, where the projection and vertical/outer-warp 4D readout kernels still become too similar after conversion into attenuation curves.

The next preflight therefore freezes the kernel-sharpening obligation before any new endpoint scoring. In a dimensionless active-window coordinate u , with $S(u)$ the smoothstep window, it proposes the source-shaped shells

$$K_{\text{proj}}^{\text{sharp}}(u) = S(u)^{1+\Pi_{\text{proj}}+C_{\text{trunc}}} \frac{1 + C_{\text{trunc}}S(u)}{1 + C_{\text{trunc}}},$$

and

$$K_{\text{vow}}^{\text{sharp}}(u) = S(u)e^{-(1+R_{\text{onset}}/R_{\text{HI}}+A_{\text{vertical}})u} + T_{\text{projected}}S(u)(1 - S(u)).$$

The first is outer-edge/truncation dominated; the second is onset/vertical-response dominated. This source-side preflight has status `SOURCE\_KERNEL\_SHARPENING\_PREFLIGHT\_READY\_NOT\_ENDPOINT`: it reduces the normalized cross-similarity of the two current kernel shapes from 0.991 to 0.644, a shape-separation gain of 0.347, without reading observed velocities or residuals. This is a formula-freeze obligation, not a fit improvement claim: V2/V3 manifests would still need to freeze these sharper kernels before any replay/holdout endpoint score is allowed.

That freeze-and-score check is now also executable. The package first freezes two source-sharpened replay formulas, with status `SHARPENED\_REPLAY\_FREEZE\_READY\_NOT\_SCORED`; this freeze gate reads only the source manifests and the sharpening preflight, and still does not score a curve. A separate scoring script then reads the frozen sharpened manifest unchanged and records `SHARPENED\_REPLAY\_HOLDOUT\_2CASE\_PRELIMINARY\_CONTROL\_RESULT`. On the same two fresh replay/holdout rows, the mean sharpened mixed RMSE is 19.224 km s^{-1} , compared with 72.919 km s^{-1} for Newtonian baryonic, 21.135 km s^{-1} for TPG/v6, 23.954 km s^{-1} for the MOND proxy, and 20.421 km s^{-1} for the exponential-disk carrier. More importantly for the Tau Core-specific question, the sharpened source-matched kernels beat all wrong-label controls in 2/2 cases, and the matched diagonal is ranked first among the two shuffled assignments. The individual matched-vs-wrong margins are -0.453 km s^{-1} for NGC5907 and -0.776 km s^{-1} for NGC7331. This does not establish population validation, but it repairs the specific failure mode exposed by the unsharpened replay/holdout audit: after source-native kernel sharpening, the correct morphology/readout lane is again distinguishable from the wrong lane on the two fresh replay rows.

A final caveat-reduction audit keeps this improvement from being overstated. It records `FOUR\_CASE\_CAVEAT\_REDUCTION\_AUDIT\_COMPLETE`: three of the four inspected-case caveats are reduced, while one is isolated but not removed. NGC5907’s prior-projection caveat is reduced to control context because the accepted mixed gate excludes the earlier projection endpoint as mixed evidence. NGC7331’s broad-window caveat is reduced for the replay path by the V2 fractional onset and source-sharpened V3 kernel, but this is not a retroactive update of the caveated accepted V1 endpoint. NGC4088’s B1 source-provenance caveat is reduced to a caveated WHISP graphical formula-freeze input, while B2 physical normalization and B3 scale uniqueness remain law-level caveats. NGC4013’s retrospective caveat is only isolated: the row remains useful frozen-reference evidence, but reducing that caveat requires a predeclared replay/holdout lane or a future source-selected analogue. No endpoint status is changed by this audit, no endpoint score is recomputed, and no population-validation claim is made.

10 Readiness and remaining blockers

Table 3: Current readiness of the Paper 8 forward endpoint.

Layer	Status
Paper 1 residual-blind disturbance association	available
Paper 2 residual-shape inference and shuffled-label null	available
Paper 3 candidate/control ladder and required- S_τ diagnostic	available
Morphology-family formula registry	defined in this package
Available-data matched Tau-proxy preflight	run; proxy only
Source-native formula preflight	diagnostic-ready
Train-selected shrinkage policy	diagnostic-ready
Quality-gate shuffled-null controls	diagnostic-ready
Endpoint decision matrix	predeclaration-ready
Predeclared endpoint protocol	ready as guardrail
Accepted source-native component tables for full sample	pending
Residual-blind morphology labels for full endpoint sample	pending
Empirical discovery claim	blocked

Thus the present package is preparation-ready, not discovery-ready. Its next scientific upgrade is not to strengthen the claim language, but to replace available-data proxy morphology observables

with accepted residual-blind morphology inputs and then run the predeclared endpoint protocol without changing gates after seeing the scores.

For that reason the repository also includes a morphology-observable intake schema. The schema records the fields required for the next run: residual-blind family labels, confidence and caveat fields, inclination and distance-quality observables, family-specific scale/support/thickness parameters, provenance, and explicit forbidden sources. In particular, it forbids using endpoint residual gain, required- S_7 , endpoint-selected radii, post-hoc family choice, or per-galaxy residual tuning as morphology inputs.

A gap audit compares the proxy manifest against this intake schema. It shows that the present package is coverage-rich but acceptance-limited: active families have broad proxy coverage, while the kernel-driving observables still need accepted residual-blind morphology sources before a discovery-style endpoint can be claimed. This makes the next upgrade concrete. The task is not to alter the endpoint gates after seeing scores, but to replace proxy morphology fields by pre-scoring observables and rerun the frozen protocol.

The repository therefore records a morphology-observable source-upgrade plan. This plan prioritizes P0 family-label, confidence, caveat, and provenance fields, then P1 active-family kernel observables such as scale, tail, compact-support, and thickness parameters. Optional ring, bar, and lopsided branches remain caveated unless external morphology or velocity-field support is supplied before scoring. The plan is a source-collection protocol, not a claim that the accepted sources have already been assembled.

Finally, the repository includes an accepted-observable manifest template and validator. The template is intentionally endpoint-blocked: it keeps the galaxy rows and pre-scoring geometry fields, but leaves accepted morphology labels, kernel observables, and provenance to be supplied from residual-blind sources. This prevents the current proxy manifest from being silently promoted into a discovery input.

An accepted-manifest readiness gate then lifts the field-level validation to an endpoint-level decision. For the current empty template, the gate returns `BLOCKED_ACCEPTED_OBSERVABLES_MISSING`. A future populated manifest must pass this gate before the frozen blind endpoint protocol is run.

A final frozen-endpoint launch guard records the same boundary operationally. In the current package it returns `LAUNCH_BLOCKED` with endpoint scores not computed. Thus the package preserves a complete path to a future blind endpoint while preventing the proxy manifest or empty accepted template from being used as discovery evidence.

The package also records an external morphology source registry for the missing accepted inputs. SPARC remains the sample and baryonic/rotation baseline; S4G is the first morphology/decomposition source; NED/NED-D supplies identity, distance, and provenance checks; DustPedia is a multiband fallback/validation layer; and PHANGS is reserved for optional high-quality non-axisymmetric or velocity-field branches. The associated SPARC crossmatch table is a to-be-filled acquisition template, not accepted-source coverage.

The first acquisition pass is included in the reproducibility package. It downloads SPARC Table 1 from the official SPARC site and S4G Pipeline 4 tables from VizieR/CDS, yielding 175 SPARC master rows, 77 S4G crossmatches, and 75 S4G/SPARC-derived disk-scale candidates. These candidates can support future source-native scale-radius fields, but they do not complete the accepted manifest: family labels, tail/ring/thickness observables, caveats, and full provenance still require the accepted-source audit path.

The repository now also builds a partial accepted morphology-observable manifest from this acquisition layer. The manifest promotes 75 S4G/SPARC-derived scale-radius observables to field-level accepted-source status, while keeping all 175 rows endpoint-blocked. This is intentional: proxy family labels still require external morphology-label audit, and scale-tail, compact, and thick/flared

branches still require their family-specific source-native kernel observables. Thus the accepted manifest draft is a data-readiness upgrade, not an endpoint score.

An accepted-manifest audit then separates audit lanes. Its closest near-term lane contains 13 exponential-disk rows with accepted S4G/SPARC scale radius and no remaining kernel-field blocker beyond the external family-label audit. These rows are not endpoint-ready; they are the next residual-blind morphology-label audit pool.

A targeted S4G component-label audit strengthens that pool. All 13 rows receive external disk-family support from S4G component decompositions; 6 rows have strict `D:expdisk` support and 7 rows remain caveated because the S4G support is barred or edge-disk. This creates a first narrow dry-run candidate lane, but endpoint scores are still not computed in the present package.

For diagnostic purposes, the package also runs an explicitly labeled narrow dry-run on this S4G-supported exponential-disk pool. The dry-run applies the accepted-scale Freeman/Bessel exponential-disk shell and records a frozen global train amplitude, shrinkage amplitudes, pool-fit diagnostic amplitudes, and a leave-one-galaxy-out all-13 stability policy. The outcome is mixed rather than confirmatory: on the 6 strict rows, the all-13 pool-fit amplitude beats TPG/v6 in 4/6 cases but beats MOND in only 2/6 cases, while the frozen global train amplitude beats TPG/v6 in 1/6 cases. The leave-one-galaxy-out all-13 policy beats TPG/v6 in 3/6 strict cases and MOND in 2/6 strict cases. This is a useful executable sanity check, not a validation result or frozen endpoint claim.

A final failure/sensitivity audit varies only the accepted scale radius by fixed multipliers 0.75, 1.0, and 1.25 under the leave-one-galaxy-out amplitude policy. The strict lane remains mixed across these multipliers, indicating that the current failure mode is unlikely to be solved by a single radius rescaling alone. The next refinement should therefore test morphology-subtype splitting or readout normalization rather than treating the scale radius as the only missing ingredient.

The package also records the corresponding inverse diagnostic: which morphology family would be inferred if the rotation-curve score table itself selected the readout family. This diagnostic is not residual-blind and is not allowed to define the accepted Paper 8 morphology labels. In the current run, the rotation-inferred family matches the predeclared proxy family in about 0.354 of rows, and matches the external S4G-supported exponential-disk label in about 0.308 of the 13 audited rows. This low agreement is not a validation result; it is a hypothesis-generation signal that the current external morphology labels and the rotation-preferred readout families may require subtype splitting.

The same diagnostic motivates a more cautious morphology-memory/history layer. Because the Tau Core bridge treats morphology as a readout structure rather than only a present-day visual class, the currently observed galaxy shape may be an insufficient proxy for the readout-relevant morphology: the galaxy may not always have had its current structure, and a solved 4D readout may encode integrated or delayed morphology/history information. The repository therefore adds a morphological memory/history proxy table. It combines source-side caveats with the inverse readout-family diagnostic and marks possible current-shape/readout-history mismatches. In the current run, the current proxy family and rotation-inferred family disagree in 113/175 rows, and 9/13 S4G-supported exponential-disk rows do not infer the exponential-disk readout family. This layer is not an accepted morphology label, not an endpoint score, and not evidence that a historical morphology has been recovered; it is a pre-endpoint hypothesis layer for future residual-blind history or morphology-memory observables.

To make that next data step operational, the repository also constructs a morphology inspection queue. The queue ranks galaxies for residual-blind image, decomposition, HI, projection, and velocity-field review, and records the requested external observables rather than changing any endpoint label. In the current run it yields 4 P0 and 18 P1 inspection targets. The top P0

cases are NGC0300, NGC6503, NGC0100, and NGC0247; all are S4G-matched current exponential-disk cases whose diagnostic readout preference points toward a scale-tail or projection-sensitive memory/history check. This queue is an acquisition plan only, not validation and not an accepted morphology manifest.

The four P0 rows are then converted into residual-blind inspection packets. Each packet records the current source context, the requested external observables, suggested catalogue lookup links, blank review fields, and forbidden inputs. The common requested observables across all four P0 packets are residual-blind multiband morphology labels, deep optical or infrared outer-disk profiles, and HI extent or asymmetry checks; NGC0247 additionally requests bar-length and velocity-field support. These packets are manual source-collection templates, not filled morphology labels.

Finally, the P0 packet layer is translated into an external imaging request manifest. This manifest records S4G-derived sky coordinates, suggested fields of view, and residual-blind source URLs for DSS2, 2MASS, WISE, NED, and SIMBAD lookups. The recommended fields of view are 8.0 arcmin for NGC0100, 29.745 arcmin for NGC0247, 20.288 arcmin for NGC0300, and 10.030 arcmin for NGC6503. This makes the next image-based morphology review operational while still leaving classification and history/memory judgments blank.

For practical review, the package also renders these four requests into a local HTML imaging-review dashboard. The dashboard is an offline launch page with per-galaxy source links, requested-observable chips, and blank review checklists. It does not embed a classification result or an accepted label; it only makes the residual-blind morphology review executable by a human reviewer.

A final source-availability preflight queries SkyView for DSS2 Red, 2MASS-K, and WISE W1 availability at the requested P0 coordinates and fields of view. All 12 P0 survey requests return at least one image candidate in the current run: 4/4 for DSS2 Red, 4/4 for 2MASS-K, and 4/4 for WISE W1. The audit intentionally records availability counts and stable query URLs only, not temporary FITS URLs and not image classifications.

The package then renders these available SkyView sources into 12 local PNG preview panels, one for each P0 galaxy–survey pair. All previews render successfully at 300 by 300 pixels. These previews are included only as review source material for a future residual-blind human morphology audit; no automated image classification, accepted label, or endpoint score is emitted.

The rendered panels are then assembled into a residual-blind visual review template. In this layer all review fields remain unfilled placeholders, and the package records the field schema needed for a future accepted morphology manifest: present-day morphology, outer-disk/LSB/tail evidence, HI asymmetry evidence, bar or $m = 2$ support, projection caveats, vertical/flare/warp evidence, compact/bulge support, ring/resonance evidence, a morphological memory/history proxy judgment, reviewer confidence, source list, and a residual-blind family recommendation. The template is not yet an accepted manifest; endpoint residual gains, required- S_τ diagnostics, best-fit readout families, MOND/RAR/TGP comparison scores, and post-hoc family switching remain forbidden inputs.

A completion gate then checks whether this visual review template has actually been filled before any accepted-manifest promotion is allowed. In the current package the gate returns `BLOCKED_VISUAL_REVIEW_PENDING`: all four P0 rows remain blocked, all 60 reviewer fields are still placeholders, accepted-manifest promotion is disallowed, and endpoint scores are not computed. This is a protocol safeguard, not a negative empirical result.

To make the blocked state actionable, the package also emits a P0 visual review handoff. The handoff lists the required residual-blind fields, local preview panels, allowed source types, and forbidden endpoint-derived inputs for each P0 galaxy. Its status is `READY_FOR_RESIDUAL_BLIND_HUMAN_REVIEW`, but it still does not promote labels or compute endpoint scores.

The handoff is finally converted into a visual-review response intake template and validator. A

Codex/source-reviewed P0 response then fills the four response rows from residual-blind source evidence: S4G Pipeline 4 components and scales, SPARC HI mass/radius evidence, DustPedia/VizieR source matches, PHANGS public-sample coverage checks, and NED/SIMBAD/SkyView lookup material. This is not a human visual review and not an endpoint calculation. It records that forbidden endpoint-derived inputs are absent: no residual gain, required- S_τ diagnostic, best-fit Tau Core readout family, MOND/RAR/TGP comparison score, or post-hoc family switching is used as a label input. The validator therefore returns `READY_FOR_INDEPENDENT_ACCEPTED_MANIFEST_AUDIT` for all four P0 rows.

A response-to-manifest promotion gate records the same boundary at the next step. For the Codex/source-reviewed P0 response it also returns `READY_FOR_INDEPENDENT_ACCEPTED_MANIFEST_AUDIT`. This does not run a frozen endpoint and does not validate Tau Core; it only allows the four P0 rows to enter a source-review audit lane. The package therefore builds a P0 Codex-source-reviewed label manifest with decision `P0_CODEX_SOURCE_REVIEW_LABELS_CREATED_NOT_ENDPOINT`. All four rows receive the residual-blind family recommendation `K_exponential_disk`, with caveats for edge projection, bar structure, or nuclear components where applicable. No full 175-galaxy endpoint labels are created and no endpoint scores are computed.

The bridge interpretation then separates this apparent 4D morphology handle from a readout-relevant morphology proxy. Thin/thick disk, bar, ring, compact, and tail labels are treated as projected 4D morphology handles rather than proven fundamental Tau-side classes. In notation, the source-review layer distinguishes K_{obs} from K_{readout} , and the formula shell is selected as $\mathcal{F}_{K_{\text{readout}}}$, not automatically as $\mathcal{F}_{K_{\text{obs}}}$. In the P0 set, the apparent 4D handle is `K_exponential_disk` for all four rows, but source-side caveats split the readout-relevant proxy layer into a clean exponential-disk control, a projection-corrected expdisk candidate, a barred/ $m = 2$ overlay candidate, and a compact-core overlay candidate. The emitted proxy table carries this distinction through `k_obs`, `k_readout`, `readout_proxy_source`, `promotion_status`, and `formula_shell`. This layer uses no rotation residuals and computes no endpoint scores.

A narrow P0 pilot then consumes only these four Codex/source-reviewed labels. Under the primary leave-one-galaxy-out all13 exponential-disk policy, the Tau readout beats TPG/v6 in 0.75 of the P0 cases and beats MOND in 0.25 of the P0 cases. The formula-shell proxy slice is stronger against TPG/v6 on this small P0 set, beating the wrong-family mean and TPG/v6 in 1.0 of cases, while still beating MOND in only 0.25 of cases. The score table records `scored_formula_shell`; in this narrow pilot it remains the direct `K_exponential_disk` control, while three P0 rows are flagged `readout_proxy_overlay_not_scored=True` because their projection, barred/ $m = 2$, or compact-core readout-proxy shell has not yet been scored. This is a useful pilot signal for the TPG-like readout direction, but it is not the frozen 175-galaxy endpoint and not empirical validation.

The package then turns the open review fields into a concrete source-acquisition plan using the requested source families: S4G, NED/NED-D, DustPedia, HI surveys, and PHANGS. This plan maps each missing P0 review field to source families, including HI extent/asymmetry checks for outer-disk and morphology-memory/history support and PHANGS/S4G support for optional bar or velocity-field evidence. All tasks remain `TO_BE_ACQUIRED_RESIDUAL_BLIND`; the layer is not an accepted morphology manifest and computes no endpoint scores.

The requested source families are then queried directly where an automated source path exists. DustPedia direct matches are found for NGC0300 only in the queried VizieR tables; PHANGS public sample coverage is not found for the four P0 galaxies, including NGC0247; and SPARC HI mass/radius evidence is present for all four P0 galaxies. These source records are useful review inputs, but they do not classify morphology and do not create accepted labels.

The package also creates a source-assisted review response draft. This draft fills the P0 response fields with residual-blind source evidence and explicit blockers, including the PHANGS no-coverage

blocker for NGC0247. It remains blocked because a human residual-blind review and independent accepted-manifest audit are still missing. It is not an accepted morphology manifest and not an endpoint score.

The next source-side test audits requested source-family availability. In the current package, S4G is partially source-ready for the P0 galaxies through existing crossmatches and disk-scale candidates, NED/NED-D lookup paths are ready, DustPedia is directly matched only for NGC0300, HI mass/radius evidence is ready for all four P0 galaxies through SPARC, and PHANGS public sample coverage is not found for the four P0 galaxies. This availability audit does not classify morphology and does not compute endpoint scores.

A targeted NGC4088 endpoint diagnostic is then added only as an interpretability check of the warp/history readout lane. This diagnostic is intentionally outside the source-bound proof chain: it scores against the observed rotation curve and therefore cannot be used as an accepted morphology label, a discovery endpoint, or a validation claim. Its purpose is narrower: to show what the already-generated warp readout candidates would look like when compared to the measured profile.

The diagnostic uses the residual-blind NGC4088 readout preflight profile as its input. That profile contains the baryonic Newtonian curve $v_N(R)$, the comparison proxy curves $\mathbf{v_v6}(R)$ and $\mathbf{v_mond}(R)$, the observed rotation curve $v_{\text{obs}}(R)$, and two source-generated Tau warp corrections $\Delta v_{\text{p1}}^2(R)$ and $\Delta v_{\text{p2}}^2(R)$. For each branch $b \in \{\text{p1}, \text{p2}\}$ and each predeclared sensitivity scenario ϵ_{cross} , the inspected Tau curve is

$$v_{\tau,b,\epsilon}(R) = \sqrt{\max(v_N^2(R) + (1 + \epsilon_{\text{cross}})\Delta v_b^2(R), 0)}.$$

The diagnostic then computes pointwise residuals $v_{\tau,b,\epsilon}(R_i) - v_{\text{obs}}(R_i)$ and reports RMSE, mean absolute error, bias, and inner/outer RMSE. The compared fixed baselines are v_N , $\mathbf{v_v6}$, and $\mathbf{v_mond}$. The fixed Tau rows use $\epsilon_{\text{cross}} = 0$; the bounded sensitivity rows use the previously derived locality interval for ϵ_{cross} . Thus the good result below is not produced by re-fitting an arbitrary curve, but it is still an endpoint diagnostic because v_{obs} is used to score and identify the best-looking bounded scenario.

Equivalently, this targeted diagnostic asks how strongly the background morphology/history layer could modulate the 4D readout in this galaxy once the leading warp candidate has been generated. The parameter ϵ_{cross} is not a new rotation-curve fit parameter; it is the bounded representative of cross-coupling between the leading warp readout and the source-side context layers, including morphology-memory, projection, and local interaction evidence. Positive ϵ_{cross} means that the background morphology layer reinforces the leading warp readout, while negative ϵ_{cross} suppresses it. The NGC4088 result therefore tests not only whether the leading warp branch has the right shape, but also whether a source-bounded morphology-context modulation can plausibly move the 4D readout toward the observed curve.

In this diagnostic, **p1** and **p2** denote two pre-generated Tau Core warp-readout candidate branches for NGC4088. Both branches descend from the same source-side warp/memory/morphology construction, but they carry different first-pass readout strength. The **p1** branch is the stronger or dominant warp-readout candidate, while **p2** is the more conservative alternative. They are not two different physical laws and they are not obtained by fitting the observed residual. They are kept separately because the bridge has not yet promoted a unique branch-selection rule.

The thick orange curve in Fig. 4 is the bounded sensitivity case

$$\text{TAU_WARP_p1_eps_+0.687500.}$$

It is the **p1** branch evaluated at the positive edge of the previously derived source-bound ϵ_{cross} locality interval. Thus it represents the strongest currently allowed positive cross-term reinforcement of the **p1** morphology readout, not an arbitrary free fit. Because the diagnostic selects the

best-looking bounded scenario after comparison with V_{obs} , this row remains endpoint-selected and diagnostic-only.

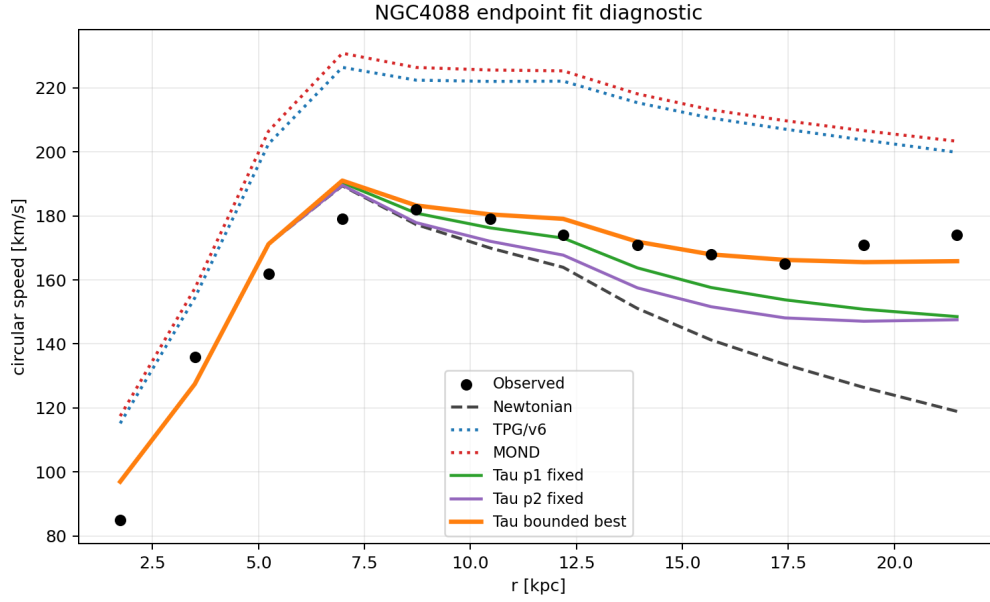


Figure 4: Endpoint diagnostic for NGC4088. The black points are the observed rotation curve. The gray, blue, and red curves are the Newtonian, TPG/v6, and MOND proxy baselines. The green and purple curves are the fixed p1 and p2 Tau Core warp-readout branches with $\epsilon_{\text{cross}} = 0$. The thick orange curve is the p1 branch at the positive edge of the source-bound ϵ_{cross} interval. This is a fit-inspection diagnostic, not an accepted endpoint or validation claim.

Numerically, on the 12-point profile the fixed p1 branch has $\text{RMSE } 12.19 \text{ km s}^{-1}$, compared with 25.40 km s^{-1} for the Newtonian proxy, 38.99 km s^{-1} for the TPG/v6 proxy, and 42.18 km s^{-1} for the MOND proxy. The bounded orange p1 sensitivity row has $\text{RMSE } 6.93 \text{ km s}^{-1}$, with most of the improvement occurring in the outer part of the curve. This is encouraging for the warp/history readout lane, but its correct claim status is still diagnostic: it motivates applying the same predeclared source-bound construction to a larger set of warp/history-rich galaxies before any empirical conclusion is drawn.

The comparison also motivates a readout-subfamily selection gate. Under the generic source-native atlas, NGC4088 is assigned to the coarse `K_thick_flared` family and the matched generic curve has $\text{RMSE } 37.81 \text{ km s}^{-1}$, worse than the Newtonian proxy. Under the targeted warp/history construction, the fixed p1 branch reaches 12.19 km s^{-1} , and the bounded ϵ_{cross} diagnostic reaches 6.93 km s^{-1} . The lesson is not that an endpoint-selected subfamily may be promoted. The lesson is that a coarse projected 4D family such as `K_thick_flared` is not necessarily the final Tau-side readout class. It may need residual-blind subfamilies such as regular thick disk, outer flare, projection-dominated thick disk, or warp/history-coupled readout. The package therefore records a `READOUT-SUBFAMILY-SELECTION-GATE`: parent-family labels remain morphology handles, while future endpoint use requires a frozen source-side rule selecting the readout-relevant subfamily before rotation scoring.

The package also builds a first-pass `READOUT-SUBFAMILY-OBSERVABLE-INTAKE` manifest. This intake lists the residual-blind fields needed to select subfamilies: warp onset or outer bend, HI asymmetry or tail support, projection safety, clean disk support, bar/core overlay support, vertical or

flare support, and morphology-history/memory support. Applied to the nine-galaxy inspection atlas, it proposes `K_disturbed_outer_tail` for the two scale-tail stress rows, `K_true_compact` for the compact rows, `K_expdisk_overlay` for the exponential-disk stress rows, `K_projection_dominated` or `K_thick_regular` for two thick/flared rows, and `K_warp_history_coupled` for NGC4088. All proposals remain proxy-only or diagnostic-only. The intake therefore does not create accepted labels; it identifies the missing source evidence needed before a future endpoint can use these subfamilies.

The corresponding source-evidence matrix shows that the bottleneck has become concrete rather than conceptual. SPARC already supplies HI mass/radius context for the disturbed-tail candidates, and S4G/SPARC supplies accepted disk-scale fields for several rows, but this is not enough to accept a readout subfamily. The disturbed-tail rows still need resolved HI asymmetry, outer-tail transition radii, and environment/history review. Compact rows need compact support radii and bulge/core decomposition evidence. Projection-dominated and thick-regular rows need projection-safety, vertical thickness, flare, or velocity-field review. NGC4088 is the closest warp/history stress case: the Tau Core source-bound protocol supplies a partial warp/history lane and an ϵ_{cross} locality bound, but accepted subfamily promotion still requires independent HI/literature confirmation of warp onset, warp asymmetry, and interaction or memory evidence. Thus the next data task is not a new endpoint fit; it is residual-blind source acquisition for subfamily acceptance.

The package therefore adds a source-citation ledger for the first readout-subfamily acquisition pass. This ledger is meant to support the future accepted-manifest review, not to promote labels. It records review-ready or candidate sources for IC2574 disturbed-gas and morphology-memory evidence [3, 14, 8], NGC4013 disk/warp overlay evidence [16], NGC5907 projection, warp, and vertical-structure evidence [10, 15, 9], NGC7331 vertical-scale evidence [7], WHISP lopsidedness/kinematic-asymmetry context for projection or overlay review [12], and UGC05716 late-type dwarf rotation/HI-context source material [11]. The accompanying local cache stores the retrieved PDFs or HTML pages where automated access succeeded; the two publisher pages that previously returned HTTP 403 are included through user-supplied local PDF copies and are marked as local-source cached. This layer is still residual-blind: it uses no endpoint residual, no required- S_r diagnostic, and no best-fit readout family.

An extraction pass then converts the cached sources into concrete residual-blind observable fields. The strongest extracted case in the current nine-galaxy inspection set is NGC5907: the source ledger records an optical warp over projected radii 13.3–24.0 kpc, maximum optical-warp displacement ~ 1.7 kpc, disk/truncation scale lengths, and independent warp/vertical-structure context. NGC7331 receives accepted vertical source fields, including molecular scale height in the ~ 100 –200 pc range and an edge-on projected HWHM of ~ 500 pc, but its low-warp classification remains caveated by possible outer-warp emission. IC2574 receives accepted disturbed-gas and morphology-memory source fields, including an extended H I envelope, H I cloud/envelope context, 48 H I holes/shells, and H I layer scaleheight ~ 350 pc, but a numeric outer-tail transition radius is still missing. NGC4013 receives source fields that pressure the current compact-only proxy toward a disk/warp overlay review, including a line-of-sight warp onset near 10 kpc and a final central H I scaleheight ~ 210 pc. UGC05716 receives H I rotation/projection context, including a kinematic inclination near 57° , but no extracted H I asymmetry or tail measurement. NGC4183 no longer remains context-only: the Verheijen–Sancisi tilted-ring source supplies a galaxy-specific 23-row orientation series over 10–241 arcsec, with $i = 82^\circ$ throughout and only a small $PA = 346^\circ$ to 349° drift. Thus the extraction pass upgrades the source layer substantially, but it still does not create broad accepted readout-subfamily labels from these inputs alone.

An accepted-manifest audit then applies the stricter promotion rule to these extracted fields. Only NGC5907 is promoted to `ACCEPTED_SUBFAMILY_SOURCE_FIELDS_ENDPOINT_STILL_BLOCKED`:

its projection-dominated source fields are complete enough for a source-field-level subfamily label, but endpoint use is still disabled. NGC7331 is `CAVEATED_ACCEPTED_SOURCE_FIELDS_NOT_ENDPOINT_READY` because the vertical fields are present while possible outer-warp emission remains unresolved. NGC4013 is `RECLASSIFICATION_REVIEW_REQUIRED`, since the extracted warp and scaleheight fields pressure the current compact-only proxy toward a disk/warp overlay review. NGC4088 is no longer empty in the extraction layer: seven warp/history-coupled fields are now recorded from the residual-blind source-bound protocol, including a first-pass 90° orientation mismatch, a 0.40 arcmin side-asymmetry proxy, a first-pass $q_{\text{warp}} = 1$, and accepted interaction/history context. However, only the interaction/history context is accepted; the warp onset, q_{warp} , memory/history proxy, and ϵ_{cross} numeric bound remain review-required or blocked. NGC4183 has now split away from the broad accepted-manifest backlog into a dedicated weak-projection/null-control lane: it does not yet promote a mixed readout-subfamily endpoint label, but it does support an accepted source-reviewed and frozen interval-control branch. IC2574 and UGC05716 remain blocked by missing required source fields, while IC4202 and UGC12506 still have no extraction rows in this pass. The audit therefore promotes source-field readiness for one galaxy only, promotes no broad endpoint labels, and computes no rotation score at the accepted-manifest stage itself.

The resulting missing-source worklist makes the next acquisition step explicit. The highest-priority case is NGC4088, where four residual-blind items block accepted subfamily use: a source-native warp-onset value, an independently reviewed q_{warp} , an accepted memory/history proxy, and a numeric ϵ_{cross} bound obtained only after the preceding two inputs are accepted. The next rows are NGC5907, which needs an endpoint-freeze rule rather than new rotation scoring, IC2574 and UGC05716, which need resolved H I tail/asymmetry measurements, and NGC4013, which needs compact-support decomposition plus reclassification review. This worklist is an acquisition guide, not a model-selection step.

The package then emits an NGC4088 source-review packet. The independent literature anchor is the Ursa Major H I synthesis data paper [13], which reports global H I geometry for NGC4088 together with a strongly distorted disk, strong position-velocity asymmetry, an asymmetric warp, a stronger position-angle change on the southern side, and NGC4085 10 arcmin to the south. These are accepted literature fields for a warp/history-coupled review. They still do not promote an endpoint-safe subfamily label: the packet marks $x_{\text{warp}} = 0.28235$ as protocol-numeric-ready but not independently accepted, keeps q_{warp} and the morphology-memory proxy in source-review-required status, accepts only the interaction/history support field, and leaves the numeric ϵ_{cross} bound blocked until q_{warp} and memory/history are accepted.

A breakthrough-oriented consolidation gate then connects this packet to the older residual-blind source-response review and coefficient-freeze layers. Under that gate, the NGC4088 warp/history chain now has six passing or caveated-passing source-bound gates: five accepted literature fields, $x_{\text{warp}} = 0.28235 \pm 0.07059$ as a dimensionally valid protocol mapping input, independently reviewed $q_{\text{warp}} = 1$ and $m_{\text{history}} = 1$, a frozen conservative $B_i = 1$ coefficient rule, and a caveated numeric protocol bound $|\epsilon_{\text{cross}}| \leq 0.6875$. The resulting status is `BREAKTHROUGH_PROTOCOL_BOUND_READY_NOT_ENDPOINT`. This is the strongest NGC4088 result in the package: it shows a continuous residual-blind path from literature morphology/warp evidence to a bounded 4D readout modulation parameter. It is not an empirical endpoint result, because the x_{warp} digitization and ϵ_{cross} bound remain protocol/caveated inputs and no accepted endpoint label is promoted.

The same source-bound logic is then applied to the cleanest projection-dominated control, NGC5907. Unlike NGC4088, this row already has a source-field-level promoted subfamily label, but it still lacked an endpoint-freeze protocol. The new projection freeze gate combines the accepted optical warp range 13.3–24.0 kpc, maximum warp displacement 1.7 kpc, disk/truncation scale lengths 4.92/4.70/1.23 kpc, interaction/warp context, vertical-structure context, and the di-

rect S4G edge-disk vertical-kernel ratio $h/R = 0.173321$. It freezes a conservative projection protocol bound $\Pi_{\text{projection}} \leq 0.789128$, with status `PROJECTION_PROTOCOL_BOUND_READY_NOT_ENDPOINT`. This gives a second mechanism-specific control lane: `NGC4088` is the warp/history source-bound stress case, while `NGC5907` is the projection-dominated source-field accepted control. Neither lane is an endpoint validation claim.

For formula inspection only, the package also runs an explicit projection-dominated endpoint diagnostic for `NGC5907`. The kernel is not fit to the observed residual. It is frozen from the projection gate as a source-windowed warp/truncation attenuation:

$$v_{\text{proj}}^2(R) = v_{\text{TPG}}^2(R) [1 - \gamma_{\text{proj}} K_{\text{proj}}(R)], \quad \gamma_{\text{proj}} = \frac{1}{2} \Pi_{\text{projection}} \frac{h}{R}.$$

Here $K_{\text{proj}}(R)$ is the smooth source-windowed projection kernel over the accepted warp/truncation range, and the sign is attenuation rather than added gravity because the projection-dominated lane is interpreted as an apparent overread/projection correction. With $\Pi_{\text{projection}} = 0.789128$ and $h/R = 0.173321$, this gives $\gamma_{\text{proj}} = 0.068386$. On the 19-point `NGC5907` curve, the diagnostic projection readout has RMSE 15.50 km s^{-1} , compared with 16.79 km s^{-1} for the TPG/v6 proxy, 18.60 km s^{-1} for the MOND proxy, 17.03 km s^{-1} for the generic promoted thick/flared proxy, and 86.48 km s^{-1} for the Newtonian proxy. This is a useful positive control: the morphology-specific projection lane improves the inspected residual structure for this galaxy. At this stage the run is only a diagnostic, because the accepted-manifest audit has not yet been paired with a formula-freeze endpoint gate.

The package then adds that missing endpoint-blind freeze audit for the same row. This audit does not recompute or improve the curve. It records, before scoring, the exact formula identifier `NGC5907\_PROJECTION\_ATTENUATION\_V1`, the attenuation sign rule, the carrier v_{TPG} , the source-windowed K_{proj} , and the amplitude rule $\gamma_{\text{proj}} = 0.5 \Pi_{\text{projection}} h/R$. The gate passes five checks: accepted source subfamily, source-frozen projection protocol, formula freeze before scoring, endpoint-blind construction, and no post-hoc retuning. Its status is `ACCEPTED\_ENDPOINT\_FREEZE\_READY`. Running the identical frozen curve after this gate gives the same numerical score, now with status `ACCEPTED\_ENDPOINT\_PRELIMINARY\_CONTROL\_RESULT`. This is the first accepted endpoint lane in the package, but only as a single-galaxy preliminary control. It is not a population validation or a universal Tau Core claim.

A final single-galaxy control audit then compares this accepted projection endpoint against wrong-family controls. The accepted `K\_projection\_dominated` readout ranks first among five family-label candidates. Its RMSE is 15.50 km s^{-1} , while the wrong-family mean is 17.63 km s^{-1} , the best wrong-family control is 16.97 km s^{-1} , and the uniform label-null family mean is 17.20 km s^{-1} . Thus the accepted projection row beats all inspected wrong-family controls for `NGC5907`. This strengthens the morphology-specific interpretation of this control case: the result is not merely that one Tau curve is good, but that the source-selected projection readout is better than the non-projection readout families in the same scoring setup. It remains a one-galaxy control audit, not a population shuffled-null result.

The same audit also preserves a negative result. The accepted `NGC5907` projection endpoint is an outer-lane readout, not a full-profile solution. Splitting the curve at the source-frozen warp/projection window gives four inner points at $R < 13.3 \text{ kpc}$, four transition points over 15.1 – 22.65 kpc , and eleven outer points at $R \geq 25.16 \text{ kpc}$. In the inner zone the mean projection kernel is exactly zero, so the accepted projection curve is identical to the TPG carrier and the RMSE remains 32.52 km s^{-1} . In the transition window the RMSE improves from 4.77 to 3.20 km s^{-1} , and in the fully active outer window it improves from 9.69 to 5.15 km s^{-1} . Therefore the positive control should be read narrowly: the projection-dominated Tau Core readout improves the source-

supported outer projection lane, while the visible inner mismatch requires a separately frozen inner disk/core/readout component before any full-profile claim can be made.

The next projection/overlay control candidate is NGC4013, but its correct status is preflight rather than endpoint. Its current manifest family is `K\compact\finite`, yet the compact lane ranks fourth in the promoted-kernel diagnostic and the source layer explicitly pressures the compact-only interpretation. The residual-blind fields include a warp/flare/disk-halo overlay flag, a line-of-sight warp onset near 10 kpc, a central H I scaleheight of 0.21 kpc, and rotational-lag context. A targeted compact/overlay source acquisition then checks the local S4G Pipeline 4 decomposition and the S4G vertical-decomposition literature. The S4G table contains an edge-disk component and a nuclear PSF for NGC4013, but no Sersic bulge component with an effective radius. The edge-disk row gives a source-native stellar $h/R = 0.232932$. Comerón et al. further describe NGC4013 as requiring thin+thick disk plus an extra extended component, with $z_{\text{EC}} \sim 3$ kpc and an extended-component mass fraction of about 0.20 [2]. Thus the compact/bulge acquisition resolves negatively: `COMPACT\_ENDPOINT\_NOT\_SOURCE\_SUPPORTED`. The updated gate status is `COMPACT\_REJECTED\_WARP\_OVERLAY\_PREFLIGHT\_READY\_FORMULA\_BLOCKED`: six gates pass or caveat-pass, and only the formula-freeze gate remains blocked. The remaining task is no longer to search for a compact radius; it is to freeze a replacement warp/vertical-overlay readout formula before any endpoint score is allowed.

The replacement formula shell is then derived, but not endpoint-frozen. The starting point is the solved-response convention

$$v_{\text{readout}}^2(R) = v_{\text{carrier}}^2(R) + \Delta v_{\text{morph}}^2(R),$$

with $v_{\text{carrier}} = v_{\text{TPG}}$ for this preflight lane. Since the source evidence points to an edge-on warp/vertical/lag overlay rather than an added-gravity tail, the residual-blind sign convention is attenuation:

$$\Delta v_{\text{morph}}^2(R) = -\Gamma_{\text{wvo}} K_{\text{wvo}}(R) v_{\text{TPG}}^2(R).$$

This gives the solved-response shell

$$v_{\text{wvo}}^2(R) = v_{\text{TPG}}^2(R) [1 - \Gamma_{\text{wvo}} K_{\text{wvo}}(R)].$$

The expression is dimensionally a velocity-squared correction because Γ_{wvo} and K_{wvo} are dimensionless. The dimensionless kernel is a source-gated mixture of the residual-blind vertical, extended-component, and lag channels:

$$K_{\text{wvo}}(R) = W_{\text{warp}}(R; R_w, R_o) \left[\omega_z K_z(R) + \omega_{\text{EC}} K_{\text{EC}}(R) + \omega_{\text{lag}} K_{\text{lag}}(R) \right],$$

where $W_{\text{warp}} = 0$ before the source warp onset and approaches 1 across the accepted outer warp window. The currently derived source fields give $R_w = 10$ kpc, $K_z(R) = 1/(1 + R/R_s)$, $K_{\text{EC}}(R) = 1/(1 + R/z_{\text{EC}})$, and $z_{\text{EC}} = 3$ kpc. The coefficient is not fitted to the observed rotation residual. Before the lag map and weights are frozen, the allowed first-order source-side bound is

$$\Gamma_{\text{wvo}} \leq \frac{1}{2} \frac{h}{R_s} + \frac{1}{2} f_{\text{EC}} = 0.216466.$$

The limits are built into the shell: if the warp window is inactive, $K_{\text{wvo}} = 0$ and $v_{\text{wvo}} = v_{\text{TPG}}$; if the overlay evidence is absent, $\Gamma_{\text{wvo}} = 0$ and the same carrier limit is recovered. A follow-up freeze gate then fills the three formula-level blockers at caveated protocol level. It sets $R_o = R_{25} = 11.2$ kpc, uses the source lag statement to define

$$K_{\text{lag}}(R) = \text{clip} \left(\frac{R_{25} - R}{R_{25} - R_{\text{lag,start}}}, 0, 1 \right), \quad R_{\text{lag,start}} = 5.8 \text{ kpc},$$

and freezes the temporary residual-blind weight rule $\omega_z = \omega_{\text{EC}} = \omega_{\text{lag}} = 1/3$. This resolves the formula-freeze blockers without using v_{obs} or endpoint residuals. It still does not permit endpoint scoring: the compact lane has been rejected, and the replacement `K\_warp\_vertical\_overlay\_candidate` label has not yet been promoted by an accepted morphology manifest. The current status is therefore `FORMULA\_FREEZE\_PROTOCOL\_READY\_LABEL\_BLOCKED`, not an accepted endpoint.

The replacement-label gate then performs that source-side promotion step. It combines the negative compact evidence, the warp/overlay pressure field, the line-of-sight warp onset, the S4G edge-disk h/R , the extended vertical component, the central H I scaleheight, and the rotational-lag context. All six promotion gates pass, with the formula-freeze gate marked caveated rather than unconditional because $R_o = R_{25}$, the lag kernel is a linear shell, and the weights are uniform until a stronger source-evidence hierarchy is derived. The resulting status is `CAVEATED\_REPLACEMENT\_LABEL\_PROMOTED\_ENDPOINT\_SCORE\_ALLOWED`. This means that a preliminary NGC4013 endpoint may now be run with the frozen warp/vertical-overlay formula, while the claim remains a caveated single-galaxy control and not a population validation.

Running that caveated endpoint gives a useful positive control. On the 36-point NGC4013 curve, the frozen warp/vertical-overlay readout has RMSE 11.45 km s^{-1} , compared with 12.27 km s^{-1} for the TPG/v6 proxy, 14.33 km s^{-1} for the MOND proxy, 16.99 km s^{-1} for the rejected compact-family readout, and 65.69 km s^{-1} for the Newtonian baryonic proxy. Thus the replacement lane improves the inspected residual structure relative to the rejected compact label and the main conventional proxies in this single case. The correct interpretation is narrow: the result supports the source-driven reclassification and formula-freeze protocol for NGC4013; it is not a population validation and remains caveated by the $R_o = R_{25}$ proxy, linear K_{lag} shell, and temporary uniform source-channel weights.

The control audit deliberately preserves a negative result. The caveated warp/vertical-overlay endpoint improves over the wrong-family mean RMSE 13.00 km s^{-1} , but it does not beat every wrong-family control: the `K\_exponential\_disk` control reaches RMSE 10.88 km s^{-1} . The matched replacement readout therefore ranks third among the five inspected family-label candidates, and the control status is `NEGATIVE\_RESULT\_MATCHED\_DOES\_NOT\_BEAT\_ALL\_WRONG\_FAMILIES`. This is scientifically useful rather than damaging: it says that the current NGC4013 warp/vertical-overlay shell is not yet a morphology-unique endpoint. Either the replacement family needs a stronger source-derived weighting/lag rule, or NGC4013 carries a smoother exponential-disk readout component that the present overlay-only formula does not yet encode.

The radial-zone audit clarifies where the positive part comes from. For the seven inner points with $R < 10 \text{ kpc}$, the frozen K_{wvo} kernel is exactly inactive and the warp/vertical-overlay curve is identical to the TPG carrier, with RMSE 12.80 km s^{-1} in both cases. In the single transition point between 10 and 11.2 kpc, the RMSE improves from 16.29 to 14.54 km s^{-1} . In the outer overlay window, the RMSE improves from 11.97 to 10.96 km s^{-1} . Thus the source-windowed readout behaves coherently as an outer-lane correction, but the wrong-family audit prevents claiming a morphology-unique solution for this galaxy.

As a diagnostic follow-up, the package tests the obvious mixed-readout hypothesis suggested by the source evidence and the wrong-family audit: a smooth exponential-disk carrier with the same frozen warp/vertical-overlay attenuation kernel. This mixed branch is explicitly not an accepted endpoint, because the exponential-disk carrier was motivated partly by the endpoint control result. Its status is `DIAGNOSTIC\_ONLY\_MIXED\_READOUT\_HYPOTHESIS`. Numerically, however, it is informative: the diagnostic mixed curve reaches RMSE 10.61 km s^{-1} , improving on the exponential-disk-only control at 10.88 km s^{-1} , the pure warp/vertical-overlay endpoint at 11.45 km s^{-1} , and the TPG/v6 proxy at 12.27 km s^{-1} . This suggests a plausible future source-blind

rule class, `K\_expdisk\_warp\_vertical\_overlay`, but that class would need to be selected from morphology/decomposition evidence before scoring before it can become an endpoint-valid readout.

A separate source-rule audit shows that this mixed interpretation is not based on intuition alone. The smooth-disk carrier has residual-blind support from the S4G edge-disk component, the available SPARC/S4G disk scales, and the absence of a Sersic bulge component. The overlay component has independent source support from the line-of-sight warp onset, vertical h/R , extended-component fraction, central H I scaleheight, and radially shallowing lag context. The compact-only lane is also explicitly rejected by the same source review. The diagnostic RMSE row is recorded but marked forbidden as a label input. The package now predeclares a general residual-blind mixed source-selection rule: a smooth carrier requires source-side disk/decomposition evidence plus a numeric disk scale, the overlay modifier requires source-side warp/projection/vertical/lag evidence plus a numeric activation or amplitude field, compact-only support must be absent or rejected, and diagnostic scores are excluded. NGC4013 passes this source rule, so the updated status is `SOURCE\_SUPPORTED\_MIXED\_HYPOTHESIS\_FORMULA\_FREEZE\_BLOCKED`. This is stronger than an intuition-based mixed label, but it is still not an endpoint result: the mixed carrier-plus-overlay formula must be frozen in a separate residual-blind gate before endpoint scoring is allowed.

That mixed formula-freeze gate is now present. It freezes the prospective mixed-readout protocol

$$v_{\text{mix}}^2(R) = v_{\text{expdisk}}^2(R) [1 - \Gamma_{\text{wvo}} K_{\text{wvo}}(R)], \quad \Delta v_{\text{mix}}^2(R) = -\Gamma_{\text{wvo}} K_{\text{wvo}}(R) v_{\text{expdisk}}^2(R),$$

where the carrier is selected by the general mixed source rule and K_{wvo} , Γ_{wvo} , the radial windows, and the source-channel weights are inherited unchanged from the frozen warp/vertical-overlay manifest. The dimensional audit passes because Γ_{wvo} and K_{wvo} are dimensionless and the correction has velocity-squared units. The carrier limits also pass: $K_{\text{wvo}} = 0$ or $\Gamma_{\text{wvo}} = 0$ recovers v_{expdisk} . The resulting freeze status is `MIXED\_FORMULA\_FREEZE\_READY\_NOT\_RETROACTIVE\_ENDPOINT`. This wording is intentional. The formula is suitable as a prospective mixed-readout protocol, but because the existing NGC4013 mixed branch was inspected after wrong-family controls, its numerical score remains diagnostic evidence rather than a retroactive accepted endpoint validation.

A separate frozen-protocol audit then reruns the same curve from the frozen manifest rather than from the earlier diagnostic branch. The construction block still does not read v_{obs} ; observed velocities enter only in the scoring block. The frozen protocol records RMSE 10.61 km s^{-1} , compared with 10.88 km s^{-1} for the exponential-disk carrier, 11.45 km s^{-1} for the pure warp/vertical-overlay readout, 12.27 km s^{-1} for TPG/v6, and 14.33 km s^{-1} for the MOND proxy. The status is `FROZEN\_PROTOCOL\_SCORE\_RECORDED\_NOT\_RETROACTIVE\_ENDPOINT`: it records that the frozen mixed protocol generates the same useful curve, while preserving that this is not an accepted endpoint validation.

The package therefore adds a population validation gate for mixed readouts. This gate states what would count as evidence rather than treating the NGC4013 result as proof. A mixed candidate must be selected from residual-blind source fields, its carrier and overlay formula must be frozen before scoring, and the matched mixed readout must beat its pure carrier, its pure overlay control, wrong-family controls, shuffled-label nulls, and the conventional baselines under the same endpoint protocol. It must also fail gracefully on non-mixed source-control galaxies; otherwise the mixed formula would be too flexible. The gate now reaches `MIXED\_POPULATION\_VALIDATION\_READY` at the protocol-preparation level: three independent prospective mixed formula protocols exist, but mixed-population endpoint scores have still not been run. The next proof step is therefore to run the predeclared mixed-vs-pure, mixed-vs-wrong, shuffled-null, baseline, and non-mixed negative-control endpoints from unchanged frozen manifests.

The package makes the preparation step executable through a mixed-readout candidate acquisition queue, with status `MIXED\_CANDIDATE\_QUEUE\_CREATED\_NOT\_ENDPOINT`. The queue

is built from source ledgers, accepted-manifest audits, SPARC/S4G scale metadata, and direct source-acquisition worklists only; it explicitly forbids v_{obs} , residuals, RMSE ranks, and required- S_τ diagnostics as selection inputs. It identifies NGC5907 as the strongest fresh formula-freeze candidate and NGC7331 as a caveated vertical/outer-warp candidate. NGC4088 and the S4G75 rows remain source-acquisition or independent-review work rather than endpoint-ready cases, while NGC4183 has now progressed into an accepted weak-projection/null-control interval-control lane.

The NGC4183 lane is important precisely because it is weak rather than strong. A residual-blind independent source review accepted the Table 4 tilted-ring transcription, preserving a near-constant $i = 82^\circ$ series and a small $PA = 346^\circ$ to 349° drift over the 23 extracted rings. The resulting source-side bound remains $\gamma_{\text{proj}} \leq 0.00269837$, implying $|\Delta v|/v \leq 0.00135010$. This was then frozen as a null-control interval rather than converted into a post-hoc point coefficient. The accepted control branch is therefore scored as an interval-control endpoint: the frozen interval is read unchanged, observed velocities enter only in the scoring block, and no residual-selected point coefficient is chosen inside the interval. Its current status is `NGC4183\_ACCEPTED\_NULL\_CONTROL\_INTERVAL\_ENDPOINT\_COMPLETE`. The correct claim boundary remains narrow: this is a single-galaxy accepted null-control interval endpoint, not a point-fit validation and not a population validation.

The NGC5907 mixed freeze gate has now been added without scoring the mixed curve. It freezes

$$v_{\text{mix}}^2(R) = v_{\text{expdisk}}^2(R) [1 - \gamma_{\text{proj}} K_{\text{proj}}(R)], \quad \Delta v_{\text{mix}}^2(R) = -\gamma_{\text{proj}} K_{\text{proj}}(R) v_{\text{expdisk}}^2(R),$$

where K_{proj} , γ_{proj} , the radial window, and the truncation contrast are inherited unchanged from the source-frozen NGC5907 projection protocol, while the carrier is the source-supported smooth exponential-disk carrier. The dimensional audit passes because K_{proj} and γ_{proj} are dimensionless and the correction has velocity-squared units. The known limits also pass: $K_{\text{proj}} = 0$ or $\gamma_{\text{proj}} = 0$ recovers v_{expdisk} . The gate status is `MIXED\_FORMULA\_FREEZE\_READY\_PRIOR\_TO\_MIXED\_SCORING`; it explicitly records `previous\_projection\_endpoint\_used\_as\_mixed\_evidence=False`.

The third prospective protocol comes from the caveated NGC7331 vertical/outer-warp lane. The source-side caveat gate maps the accepted molecular scale-height range 100–200 pc, projected edge-on HWHM ~ 500 pc, and SPARC disk scale $R_s = 5.02$ kpc into dimensionless vertical handles, giving an intrinsic midrange $h/R_s \simeq 0.0299$ and projected HWHM/ $R_s \simeq 0.0996$. The V1 formula was frozen with the outer-warp evidence treated as context-only rather than as a numeric onset, so the frozen formula is explicitly caveated:

$$v_{\text{mix}}^2(R) = v_{\text{expdisk}}^2(R) [1 - \gamma_{\text{vow}} K_{\text{vow}}(R)].$$

The radial window is a broad source-scale outer window from R_s to R_{HI} , not a claimed measured warp-onset radius. A later source-only tightening gate records a fractional Bosma onset for NGC7331, $R_{\text{onset}} \simeq 0.5 R_{\text{Ho}}$, together with the Holmberg-system major-axis size $13.5'$ [1]. Combined with the SPARC distance this gives an approximate candidate $R_{\text{onset}} \simeq 14.43$ kpc, or $2.87 R_s$. This does not alter the already scored V1 endpoint; it defines a replay/holdout-only V2 window. The package therefore also freezes `NGC7331\_FRACTIONAL\_ONSET\_V2\_REPLAY\_FREEZE\_V1`, with the same carrier and amplitude rule but with the V2 outer window 14.43 kpc $\rightarrow R_{\text{HI}}$. Its status is `V2\_REPLAY\_PROTOCOL\_READY\_NOT\_SCORED`; the V1 status remains `CAVEATED\_MIXED\_FORMULA\_FREEZE\_READY\_PRIOR\_TO\_MIXED\_SCORING`. With NGC4013, NGC5907, and caveated NGC7331, the package reaches three frozen prospective mixed protocols.

The predeclared mixed-population endpoint is then run from these unchanged manifests. Its status is `MIXED\_POPULATION\_ENDPOINT\_PRELIMINARY\_CONTROL\_RESULT`. Across the three

cases, the mean RMSE of the mixed readout is 16.41 km s^{-1} , compared with 70.51 km s^{-1} for Newtonian baryonic, 18.18 km s^{-1} for the TPG/v6 proxy, 20.75 km s^{-1} for the MOND proxy, and 17.24 km s^{-1} for the exponential-disk carrier. The mixed readout improves over each of these four comparators in all three rows. This is a useful endpoint signal, but not yet a population-validation claim: NGC4013 remains a retrospective frozen-reference protocol, NGC7331 keeps the broad-window caveat, and no wrong-family or shuffled-label population null has yet been run for this three-case mixed endpoint.

A first wrong-label control audit is then run on the same frozen packet. Each of the three frozen mixed formula shells is applied to each of the three galaxies, and all $3!$ one-to-one formula-label assignments are scored. The source-matched diagonal has mean RMSE 16.41 km s^{-1} , while the wrong-label mean is 17.29 km s^{-1} . In each galaxy, the matched frozen formula ranks first among the three mixed formula labels. The matched diagonal is also the best of the six one-to-one assignments, with the best shuffled assignment at 16.71 km s^{-1} . The audit status is `PASSED\_3CASE\_WRONG\_LABEL\_AND\_SHUFFLED\_CONTROL`. This is the first direct support for the intended Tau Core claim that source-matched morphology readouts can matter, but it remains a small- N control audit rather than a population validation.

The NGC5907 mixed row is then promoted one step further as its own accepted single-galaxy mixed endpoint. The accepted gate records `ACCEPTED\_MIXED\_ENDPOINT\_FREEZE\_READY`: the mixed formula was frozen before scoring, the construction remains endpoint-blind, no post-score retuning is allowed, and the prior NGC5907 projection endpoint is explicitly not used as mixed-readout evidence. The separate scoring script then reads the frozen manifest unchanged and records `ACCEPTED\_MIXED\_ENDPOINT\_PRELIMINARY\_CONTROL\_RESULT`. On the 19-point NGC5907 curve, the accepted mixed readout has RMSE 16.37 km s^{-1} , compared with 16.79 km s^{-1} for the best local baseline, TPG/v6. It also beats the inspected wrong mixed-family controls: the wrong-mixed mean is 17.06 km s^{-1} , and the best wrong mixed control is 16.85 km s^{-1} . Thus NGC5907 is no longer only a row inside the three-case mixed packet; it is also an accepted single-galaxy mixed control endpoint. This still does not make a population-validation claim.

The NGC4013 mixed row is treated more strictly. A dedicated blocker gate records `MIXED\_ACCEPTED\_ENDPOINT\_BLOCKED\_RETROACTIVE\_PROTOCOL\_READY`. The source rule is ready, the formula is frozen for future scoring, and the construction is endpoint-blind; the frozen mixed protocol has RMSE 10.61 km s^{-1} , better than the best local baseline at 10.88 km s^{-1} and better than the inspected wrong mixed-family controls. Nevertheless, the existing score is not promoted to an accepted endpoint because the mixed exponential-disk plus WVO lane was developed after the NGC4013 wrong-family/control context had already been inspected. This is a negative protocol result that is deliberately preserved: the formula may be used prospectively or in a predeclared replay/holdout lane, but the current NGC4013 mixed score remains a retrospective frozen-reference row.

The follow-up retrospective-caveat closure gate narrows this remaining caveat without erasing it. Its status is `NGC4013\_RETROSPECTIVE\_CAVEAT\_CLOSURE\_PATH\_FORMALIZED\_NOT\_CLOSED`: the source rule is transferable, the frozen formula is transferable, and the control signal is recorded, but the retrospective score remains forbidden as accepted endpoint evidence. The gate also records that there is currently no exact non-NGC4013 analogue with the same source-rule readiness; NGC4088 is only the nearest source-bound analogue and its formula-freeze route is not the same protocol. Thus the NGC4013 caveat is now operational rather than vague: it can be reduced only by a predeclared replay/holdout lane or by a future source-selected analogue, not by relabeling the already inspected score.

The predeclared replay/holdout gate then turns that operational caveat into explicit future-use rules. Its status is `NGC4013\_PREDECLARED\_REPLAY\_HOLDOUT\_GATE\_BUILT\_ENDPOINT`

`_STILL\_BLOCKED`. The source rule and frozen `expdisk+warp/vertical-overlay` formula are transferable, the prospective protocol is ready, and the existing score is quarantined. The gate explicitly blocks a same-curve replay, because the NGC4013 curve and controls have already been inspected. It allows only future routes: a genuinely uninspected predeclared holdout target, a source-selected non-NGC4013 analogue, or a predeclared population replay packet. Since the exact non-NGC4013 analogue count remains zero, this narrows the caveat but does not close it.

The remaining-caveat roadmap then ranks the next admissible caveat-reduction step without changing any endpoint score. After the dedicated NGC7331 replay described below, the NGC4013 predeclared replay/holdout gate, and the NGC4088 remaining-caveat action gate are included, its status is `REMAINING\_CAVEAT\_CLOSURE\_ROADMAP\_UPDATED\_AFTER\_NGC4088\_ACTION\_GATE\_NOT\_ENDPOINT`. Among the four inspected cases, no route remains replay-ready without new source acquisition, one replay path has been completed without updating the accepted V1 endpoint, one predeclared replay gate has been built without endpoint permission, and one remaining-caveat action gate has been built. The next recommended gate is `B2\_CLOSURE\_FUNCTIONAL\_AND\_ASYMPTOTIC\_CARRIER\_DERIVATION\_FOR\_NGC4088`. This does not change the NGC4088 endpoint status; it identifies the next theory task needed to turn the strongest single-galaxy control result into a more internally derived Tau Core readout.

That replay/holdout endpoint is now run as a separate scoring script. Its status is `NGC7331\_V2\_V3\_REPLAY\_HOLDOUT\_PRELIMINARY\_CONTROL\_RESULT`. The accepted V1 broad-window reference remains unchanged at $\text{RMSE } 22.26 \text{ km s}^{-1}$. The V2 fractional-onset replay row gives $\text{RMSE } 22.73 \text{ km s}^{-1}$, while the V3 source-sharpened replay row improves to 22.13 km s^{-1} . V3 also beats the best local baseline, the exponential-disk carrier at 23.47 km s^{-1} , and the wrong sharpened projection control at 22.91 km s^{-1} . This reduces the NGC7331 broad-window caveat for the replay path, but it is not a retroactive update of the already accepted V1 endpoint and not a population-validation claim.

The caveated NGC7331 mixed row is also promoted, but only with its caveat attached. Its accepted gate records `CAVEATED\_ACCEPTED\_MIXED\_ENDPOINT\_FREEZE\_READY`: the vertical/outer-warp formula was frozen before scoring, the construction remains endpoint-blind, and the V1 source layer records that the endpoint used a broad window. The separate scoring script reads the frozen manifest unchanged and records `CAVEATED\_ACCEPTED\_MIXED\_ENDPOINT\_PRELIMINARY\_CONTROL\_RESULT`. On the 36-point NGC7331 curve, the caveated accepted mixed readout has $\text{RMSE } 22.26 \text{ km s}^{-1}$, compared with 23.47 km s^{-1} for the best local baseline, the exponential-disk carrier. It also beats the inspected wrong mixed-family controls: the wrong-mixed mean is 22.67 km s^{-1} , and the best wrong mixed control is 22.67 km s^{-1} . This is a narrow positive control, not a removal of the caveat: the scored radial window remains the V1 broad source-scale outer window from R_s to R_{HI} . The V2 replay freeze manifest gives a candidate inner window near 14.43 kpc , but using it requires a predeclared replay/holdout run rather than retroactively changing this endpoint.

The expansion gate then asks which galaxy can be added next without using the already inspected endpoint residuals. It returns `NEXT\_MIXED\_CASE\_IDENTIFIED\_FORMULA\_FREEZE\_BLOCKED`. The closest next candidate is NGC4088, whose source-bound warp/history chain already contains $x_w \simeq 0.282$, $q_{\text{warp}} = 1$, $m_{\text{history}} = 1$, and $|\epsilon_{\text{cross}}| \leq 0.6875$. However, NGC4088 is not promoted to a fourth mixed endpoint. Its formula freeze remains blocked by independent digitization review, physical normalization-law derivation, and scale-uniqueness: the package records multiple residual-blind candidate scales for the same warp/history lane. Thus the expansion result is useful precisely because it prevents the strong NGC4088 diagnostic curve from becoming endpoint evidence before its source-side formula gate is closed.

The follow-up blocker-resolution ledger gives this decision a more operational form, with status

NGC4088_FORMULA_FREEZE_BLOCKER_RESOLUTION_PLAN_CREATED. It separates three local formula-freeze blockers from one population-claim blocker. The local blockers are: an independent residual-blind review or reproducible image-analysis repeat for the source-native x_w digitization; a Tau-side physical normalization law mapping the dimensionless closure-source basis into Δv^2 ; and a unique residual-blind scale-selection rule, since five dimensionally valid candidate scales are recorded for the same lane. Population generalization is tracked separately: it blocks broad claims, but it does not replace the local freeze obligations. The ledger also records three protocol-ready supports, including the breakthrough warp/history chain, the accepted $q_{\text{warp}}/\text{memory}/B_i$ protocol-bound review, and the expansion-gate priority. None of these supports authorizes endpoint scoring by itself.

The first blocker is then narrowed to an executable review packet rather than left as an informal caveat. The package emits INDEPENDENT_XW_REVIEW_PACKET_READY_RESPONSE_PENDING. This freezes the second-review route for the NGC4088 channel-map digitization: the reviewer or frozen image-analysis repeat may use only the page-76 region of interest, the worksheet overlay, and the page-77 cross-check, and may not use v_{obs} , rotation residuals, endpoint scores, fit ranks, or required- S_τ diagnostics. The current first-pass value is $x_w = 0.28235$, with an acceptance tolerance $\Delta x_w = 0.07059$. This does not resolve the blocker. It only defines how the blocker can be resolved: an independent onset measurement must either reproduce the first-pass value within the frozen tolerance or force a source-side uncertainty interval before any formula freeze.

The package also runs a frozen image-analysis repeat attempt on the same page-76 channel-map region. This repeat extracts dark-contour component geometry from 23 frozen worksheet panels and finds elongated components in 21 panels, with a warp-like position-angle departure signal. The result is useful support for the review path, but its status is FROZEN_IMAGE_REPEAT_ATTEMPT_COMPLETE_INCONCLUSIVE. It does not accept a new x_w , because the printed channel-map crop still lacks a robust machine-readable source-native radial calibration and the contour-component selection is not strong enough to replace independent review. Thus B1 remains open.

The radial-calibration packet makes that failure mode explicit. It records the valid first-pass conversion $x_w = R_{\text{warp}}/R_{\text{HI}} = 0.28235$, but gives the calibration status RADIAL_CALIBRATION_NOT_ACCEPTED. The acceptable closure routes are now predeclared: an independent reviewer can report source-native side onsets in arcminutes, the frozen image repeat can be upgraded with radial tick or coordinate calibration, or an original source-coordinate H I product can be used. Until one of these routes closes, the first-pass x_w remains a preflight mapping input rather than a formula-freeze input.

The original-data route is then audited as its own reproducible source-acquisition step. The package probes the WHISP catalog route, direct UGC7081/NGC4088 pages, cached printed-page context, and a NED metadata gateway, without using v_{obs} , endpoint residuals, or score ranks. Its current status remains RC3_ORIGINAL_CHANNEL_MAP_DATA_ROUTE_AUDITED_NO_DIRECT_PRODUCT_CACHED: no direct source-native H I/FITS product has yet been cached and accepted for the B1 formula-freeze route. However, the corrected WHISP identifier UGC7081 does yield a source-provenance graphical H I overview and observation/reduction notes. These are cached and converted into the review packet WHISP_OVERVIEW_EXTRACTION_PACKET_READY_RESPONSE_PENDING. The position-velocity panel includes an “Offset from center (arcmin)” axis, so it can support a residual-blind direct-arcmin onset review. A frozen saturation-component extraction is then run on that panel. It finds two opposite-side coherent components and gives $x_{w,\text{review}} = 0.29833$, which agrees with the first-pass $x_w = 0.28235$ within the frozen tolerance. Its status is FROZEN_WHISP_OVERVIEW_EXTRACTION_ATTEMPT_COMPLETE_AGREES_WITH_FIRST_PASS. A subsequent promotion review records B1_CAVEATED_XW_ACCEPTED_FOR_FORMULA_FREEZE_NOT_ENDPOINT: the WHISP overview extraction is accepted as the B1 x_w formula-freeze

input, with an explicit provenance caveat that it comes from a graphical overview rather than a direct source-coordinate H I product. Thus B1 is closed for formula-freeze bookkeeping, but this does not allow endpoint scoring and the caveat must travel with any later NGC4088 formula.

The second blocker is also narrowed without being prematurely closed. The B2 synthesis status is `B2\_FORMULA\_CONDITIONAL\_DERIVATION\_SYNTHESIZED\_LAW\_STILL\_OPEN`. The current NGC4088 warp/history normalization can be written as

$$\delta v_{\text{warp}}^2(R; p) = \lambda_w C_{\text{warp}}(R/R_{\text{HI}}; x_w, p), \quad \lambda_w = \sigma_{\text{warp}} q_{\text{warp}} x_w V_{\text{flat}}^2,$$

and the numerical candidate is dimensionally valid, residual-blind, and executable. After the WHISP graphical-overview review is promoted, the B2 synthesis is aligned to the accepted formula-freeze manifest: the endpoint formula uses $x_w = 0.298333$ and hence $\lambda_w = 8795.111752 \text{ km}^2 \text{ s}^{-2}$. The earlier $8324.016 \text{ km}^2 \text{ s}^{-2}$ value is retained only as first-pass provenance from the pre-review x_w , not as the frozen endpoint normalization. The synthesis combines the scale-selection, asymptotic-carrier, closure-functional, target-origin, Euler-ansatz, separability, and cross-term gates. Its verdict is deliberately conservative: the formula shell is strong enough to state, but the Tau-side physical normalization law is not derived. The remaining law obligations are a closure functional $J_\tau[\lambda_w]$, an asymptotic-carrier theorem selecting V_{flat}^2 , a separability or cross-term bound for the product form, and population transfer for broader claims. Thus B2 is not closed as a final physical law, but it can be closed at protocol formula-freeze level if the caveat is carried forward.

The third blocker is then narrowed to a conditional uniqueness result. The B3 synthesis status is `B3\_CONDITIONAL\_UNIQUE\_SCALE\_SELECTED\_LAW\_LEVEL\_UNIQUENESS\_OPEN`. The original scale audit records five dimensionally valid, residual-blind candidate scales. Under the frozen `MINIMAL\_SOURCE\_ONSET\_ASYMPTOTIC\_CARRIER\_RULE`, only `CURRENT\_XW\_VFLAT2` survives. This resolves the multiple-scale ambiguity at protocol level: the alternatives fail because they use point-sampled baryonic medians, external TPG-like comparator carriers, or additional closure-fraction factors not derived by the Tau-side rule. However, law-level scale uniqueness remains open, because the rule itself depends on the B2 asymptotic-carrier and closure-functional obligations. B3 is therefore conditional-unique but still endpoint-blocking.

The follow-up warp/history formula-freeze gate then freezes the first executable NGC4088 protocol shell without scoring it. It uses the caveated WHISP graphical-overview value $x_w = 0.298333$, $q_{\text{warp}} = 1$, the positive source-side warp/history sign, $V_{\text{flat}} = 171.7 \text{ km s}^{-1}$, and the minimal linear onset branch $p = 1$. The frozen formula is

$$v_{\text{readout}}^2(R) = v_{\text{Newt}}^2(R) + \lambda_w C_{\text{warp}}(R/R_{\text{HI}}; x_w, p), \quad \lambda_w = \sigma_{\text{warp}} q_{\text{warp}} x_w V_{\text{flat}}^2 = 8795.111752 \text{ km}^2 \text{ s}^{-2}.$$

Here $C_{\text{warp}} = q_{\text{warp}} \max(0, (x - x_w)/(1 - x_w))^p$ is dimensionless, so the dimensional audit passes; the inactive-window and zero-source limits recover the Newtonian baryonic carrier. The quadratic $p = 2$ branch is kept only as a sensitivity/control branch, not as the endpoint formula. The gate status is `NGC4088\_WARP\_HISTORY\_FORMULA\_FREEZE\_READY\_LAW\_CAVEATED\_NOT\_SCORE`: B2 is closed only as a formula-conditional protocol, and B3 is closed only as protocol-level scale uniqueness.

The follow-up B2 conditional-derivation gate then records the strongest current law-side statement without promoting it to a final law. Its status is `B2\_CONDITIONAL\_THEOREM\_ALIGNED\_TO\_FREEZE\_MANIFEST\_LAW\_PREMISES\_OPEN`. A source-load closure-functional subgate constructs the explicit conditional functional

$$J_{\text{load}}[\lambda_w] = \frac{1}{2} \kappa_\lambda \|(\lambda_w - \Lambda_\tau) C_{\text{warp}}\|_W^2, \quad \Lambda_\tau = \sigma_{\text{warp}} q_{\text{warp}} x_w V_{\text{flat}}^2.$$

For $\kappa_\lambda > 0$ and nonzero kernel norm, its Euler condition gives $\lambda_w = \Lambda_\tau$, aligned with the frozen $8795.111752 \text{ km}^2 \text{ s}^{-2}$ normalization. A source-load-origin gate then decomposes Λ_τ using the formula-freeze manifest. It records `SOURCE\_LOAD\_ORIGIN\_PARTIALLY\_GROUNDED\_CARRIER\_AND\_CROSS\_OPEN`: freeze alignment, dimensions/limits, conditional stationarity, and caveated x_w source acceptance pass; the sign/strength law is formula-conditional; and the cross-term layer is only partial because the sharp B_i coefficient rule is ready but the residual-blind ϵ_{cross} source bound is still incomplete. A frozen asymptotic-carrier theorem gate further sharpens the V_{flat}^2 premise to `FROZEN\_VFLAT2\_CARRIER\_CONDITIONAL\_THEOREM\_LAW\_PROOF\_OPEN`: under the residual-blind, source-native, asymptotic, non-comparator, minimally factorized onset rule, the frozen protocol carrier is V_{flat}^2 . This is still not a proof of the final Tau-side carrier law, because alternative-carrier exclusion and population transfer remain open. Overall, the B2 result is stronger than an informal ansatz because the functional, stationarity equation, source-load decomposition, carrier criterion, dimension check, zero-source limit, and inactive-window limit are explicit. It is still not a proof of the Tau-side physical law: the final carrier theorem, the sign/strength source-load law, and the separability/cross-term proof remain open.

A population-transfer preflight gate then checks whether this B2 carrier theorem is already a population result. Its status is `POPULATION\_TRANSFER\_PREFLIGHT\_BUILT\_EXACT\_TRANSFER\_BLOCKED\_ANALOGUE\_LANE\_AVAILABLE`. The gate finds one exact reference protocol, NGC4088; no independent exact transfer-ready warp/history case; three useful but non-identical mixed overlay/projection analogues, NGC4013, NGC5907, and NGC7331; and three blocked acquisition controls. This is an important narrowing of the claim: the analogues support a broader readout-program direction, but they do not yet transfer the exact $\sigma_{\text{warp}} q_{\text{warp}} x_w V_{\text{flat}}^2$ source-load law. A population claim requires independent exact warp/history cases whose x_w , q_{warp} , σ_{warp} , V_{flat} , and cross-term source observables are frozen before scoring.

The corresponding exact-transfer candidate manifest makes this input contract explicit. It records five required source-side fields, x_w , q_{warp} , σ_{warp} , V_{flat} , and ϵ_{cross} -bounding observables, and returns `EXACT\_TRANSFER\_CANDIDATE\_MANIFEST\_BUILT\_NO\_READY\_INDEPENDENT\_CASE`. In that manifest, NGC7331 is the closest current upgrade target because it already has a residual-blind fractional outer-warp onset candidate, but it lacks exact q_{warp} , sign-rule, and cross-term source observables. A dedicated NGC7331 upgrade gate records this as `NGC7331\_EXACT\_TRANSFER\_UPGRADE\_GATE\_BUILT\_FORMULA\_FREEZE\_BLOCKED`: x_w and V_{flat} are available, while q_{warp} , σ_{warp} , and ϵ_{cross} remain blocked. A follow-up source packet now turns those three blockers into explicit residual-blind measurement and review templates, with status `NGC7331\_EXACT\_TRANSFER\_SOURCE\_PACKET\_BUILT\_MEASUREMENTS\_PENDING`. A source-evidence review then records `NGC7331\_SOURCE\_EVIDENCE\_REVIEW\_BUILT\_EXACT\_TRANSFER\_STILL\_BLOCKED`: the complex H I warp context makes NGC7331 scientifically relevant for transfer, but it also means q_{warp} , σ_{warp} , and ϵ_{cross} cannot be inherited from NGC4088. A source-acquisition route has now cached and FITS-audited the THINGS H I moment maps and records `NGC7331\_THINGS\_QWARP\_WORKSHEET\_READY\_MEASUREMENT\_PENDING`; Bosma/NED figure digitization remains the fallback. A first-pass THINGS measurement then gives a residual-blind source-native q_{warp} candidate, but keeps it review-required: the conservative ridge-centroid observable gives $q_{\text{warp}} \simeq 0.0068$, while a threshold-sensitivity audit finds a stable outer-envelope p80 range $q_{\text{warp}} \simeq 0.195\text{--}0.217$. A MOM1 sign/cross-term review adds kinematic context: the natural and robust maps agree on the receding side, preserve the inner/outer receding orientation, and have small morphology-kinematic PA mismatch ($f_{\text{PA,max}} \simeq 0.043$), but the sign is still not frozen and the candidate ϵ_{cross} bound ($\simeq 0.489$) remains review-required. A dedicated observable-choice gate then records `NGC7331\_QWARP\_OBSERVABLE\_CHOICE\_REVIEW\_GATE\_BUILT\_FREEZE\_BLOCKED`: the midpoint envelope observable is about 25.9 times the midpoint centroid observable, so the

bridge cannot honestly freeze q_{warp} by selecting one observable by fiat. A follow-up review packet freezes the allowed outcomes `ACCEPT\_CENTROID`, `ACCEPT\_ENVELOPE`, `CARRY\_INTERVAL`, and `REJECT\_Q\_FREEZE`, while explicitly forbidding rotation residuals, endpoint scores, baseline RMSE, wrong-family ranks, and required- S_r diagnostics as review inputs. A source-only response then selects `CARRY\_INTERVAL`: it carries $q_{\text{warp}} \in [0.0079404475812108, 0.2057957876154617]$, carries MOM1 orientation context, and carries the conservative ϵ_{cross} candidate bound $\simeq 0.489$, using only allowed THINGS source files and no endpoint information. The corresponding intake validator now returns `NGC7331\_QWARP\_REVIEW\_INTAKE\_FORMULA\_FREEZE\_INPUT\_READY\_NOT\_ENDPOINT`. This removes the response-pending blocker, but it is still formula-freeze preparation rather than an endpoint score: the next NGC7331 exact-transfer gate must freeze the actual formula while carrying the q_{warp} interval and cross-term caveat.

After this response/intake step, the regenerated NGC7331 exact-transfer upgrade gate records `NGC7331\_EXACT\_TRANSFER\_UPGRADE\_FORMULA\_FREEZE\_INPUT\_READY\_NOT\_ENDPOINT`: x_w , V_{flat} , q_{warp} , sign context, and ϵ_{cross} inputs are all available for formula-freeze preparation, with zero blocked source-input gates, but endpoint scores remain disallowed. The resulting exact-transfer freeze gate records `NGC7331\_EXACT\_B2\_TRANSFER\_INTERVAL\_FORMULA\_FREEZE\_READY\_NOT\_SCORE`. It freezes the NGC4088-style B2 shell as an interval protocol: $q_{\text{warp}} \in [0.0079404475812108, 0.2057957876154617]$, $\lambda_w \in [242.344646, 6280.944095] \text{ km}^2 \text{ s}^{-2}$, and a cross-term-caveated interval $[123.941984, 9349.633732] \text{ km}^2 \text{ s}^{-2}$. The gate deliberately preserves the current B2 convention in which q_{warp} appears both in λ_w and in C_{warp} , and labels that placement as a law-level caveat. The freeze grid contains only radius/carrier and source-frozen interval columns; it is not an endpoint score.

The frozen interval can then be audited as an interval, still without choosing a residual-selected q_{warp} point. This post-freeze control audit records `NGC7331\_EXACT\_B2\_INTERVAL\_CONTROL\_AUDIT\_COMPLETE\_NOT\_POINT\_ENDPOINT`. On the 36 NGC7331 points, both the strict and cross-caveated exact-transfer intervals have zero coverage of the observed rotation curve, with distance-to-interval RMSE 56.27 km s^{-1} and 54.81 km s^{-1} , respectively. The best local point baseline in this audit is the exponential-disk carrier, with RMSE 23.47 km s^{-1} . This is therefore a weak or negative exact-transfer result for NGC7331, not an accepted point endpoint. Its role is nevertheless important: it shows that a source-side interval freeze alone does not license population transfer of the NGC4088 B2 shell, and that any future NGC7331 point-branch or alternative source-derived readout must be predeclared before scoring.

A consolidated readiness dashboard then combines these three local blockers into the single status `NGC4088\_FORMULA\_FREEZE\_READINESS\_DASHBOARD\_CREATED`. The dashboard records 3/3 local formula-freeze blockers resolved: B1 is closed with the caveated WHISP graphical-overview x_w , B2 is `B2\_PROTOCOL\_FORMULA\_FREEZE\_READY\_LAW\_LEVEL\_OPEN`, and B3 is `B3\_PROTOCOL\_UNIQUE\_SCALE\_SELECTED\_LAW\_LEVEL\_OPEN`. Its readiness decision is therefore `FORMULA\_FREEZE\_READY\_ENDPOINT\_GATE\_REQUIRED`: NGC4088 now has a frozen prospective formula protocol, but it is still not an endpoint score and not an empirical validation result. Any endpoint run must be a separate accepted-gate script that reads the frozen manifest unchanged.

That accepted-gate script is then run as a caveated single-galaxy control. The gate status is `CAVEATED\_ACCEPTED\_ENDPOINT\_FREEZE\_READY`: the formula was frozen before scoring, construction remains endpoint-blind, and the B1/B2/B3 caveats are explicitly attached. The separate scoring script reads v_{obs} only after the gate passes. It records `CAVEATED\_ACCEPTED\_ENDPOINT\_PRELIMINARY\_CONTROL\_RESULT`. On the 12-point NGC4088 curve, the frozen warp/history readout has RMSE 11.62 km s^{-1} , compared with 25.40 km s^{-1} for Newtonian baryonic, 38.99 km s^{-1} for TPG/v6, and 42.18 km s^{-1} for the MOND proxy. It also beats the inspected wrong-family Tau controls: the wrong-family mean is 41.86 km s^{-1} , and the best wrong-family control, `K\_exponential`

`_disk`, is 37.87 km s^{-1} . Thus the matched frozen NGC4088 readout ranks first among all inspected models in this endpoint. This is the strongest NGC4088 result in the package, but its claim boundary remains narrow: it is a caveated single-galaxy preliminary control endpoint, not a population validation and not a closure of the B2/B3 law-level Tau-side derivations.

The NGC4088 remaining-caveat action gate then separates endpoint status from the remaining provenance and theory obligations. Its status is `NGC4088\_REMAINING\_CAVEAT\_ACTION\_GATE\_BUILT\_NOT\_ENDPOINT`. The endpoint is already caveated-control scored, and B1 is closed for formula freeze using the WHISP graphical-overview x_w , while the direct H I provenance upgrade is useful but optional rather than endpoint-blocking. The primary remaining theory caveat is B2: the physical normalization is now algebraically derived from a conditional source-load closure functional, but it remains formula-conditional until the Tau-side source-load origin, the asymptotic-carrier theorem, and the cross-term/separability proof are derived. B3 is then dependent on that result, because law-level scale uniqueness requires the same carrier/closure derivation. The next scientific action is therefore B2 source-load-origin and asymptotic-carrier work, not another NGC4088 endpoint score.

The package therefore now contains four inspected endpoint/readout cases, but they should not be conflated into a uniform four-galaxy population validation. The consolidated status is `FOUR\_INSPECTED\_CASES\_HETEROGENEOUS\_PRELIMINARY\_EVIDENCE`: NGC4013, NGC5907, and caveated NGC7331 form the three-case mixed-readout control packet; NGC5907 and caveated NGC7331 are additionally promoted to accepted mixed single-galaxy endpoint rows inside that packet; and NGC4088 is an additional caveated single-galaxy warp/history endpoint. Across these four inspected cases, the source- or morphology-matched Tau Core readout beats the best local baseline in 4/4 rows and also beats the inspected wrong-family controls in 4/4 rows. Three rows now have accepted single-galaxy endpoint status: NGC5907 as an accepted mixed control endpoint, NGC7331 as a caveated accepted mixed control endpoint, and NGC4088 as a caveated accepted warp/history endpoint. This is encouraging evidence for readout specificity, not empirical validation of the theory: the evidence packets are heterogeneous, NGC4013 remains a retrospective frozen-reference row with its closure path formalized but not closed, NGC7331's scored V1 row keeps its broad-window caveat even though a later source-only gate identifies a replay-only fractional warp-onset candidate, and NGC4088 still carries the B1 graphical-overview provenance caveat plus B2/B3 law-level open obligations.

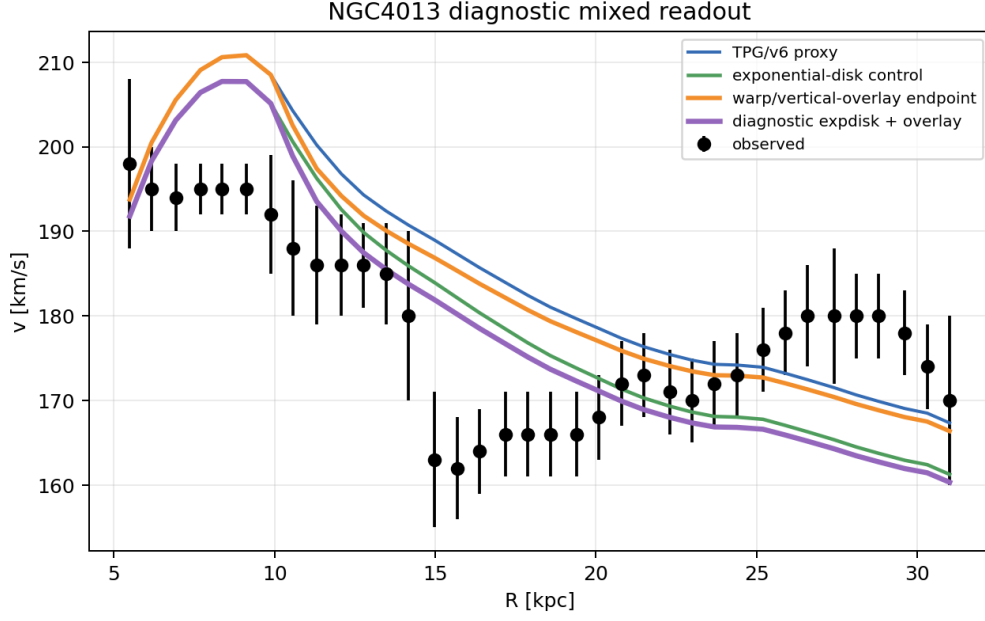


Figure 5: Diagnostic mixed readout for NGC4013. The purple curve applies the frozen warp/vertical-overlay attenuation kernel to the exponential-disk carrier. It improves the inspected RMSE, but it is not an accepted endpoint because the mixed carrier was motivated by the wrong-family control audit.

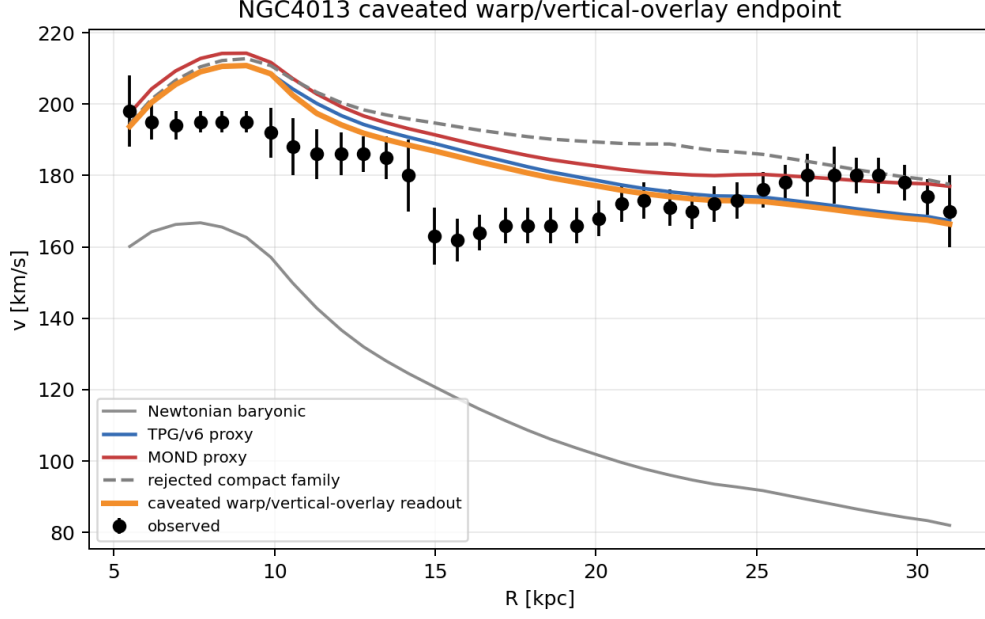


Figure 6: Caveated warp/vertical-overlay endpoint for NGC4013. The black points are the observed rotation curve. The gray, blue, and red curves are the Newtonian, TPG/v6, and MOND proxy baselines. The dashed gray curve is the rejected compact-family Tau readout. The orange curve is the source-frozen warp/vertical-overlay replacement readout $v_{\text{wvo}}^2 = v_{\text{TPG}}^2(1 - \Gamma_{\text{wvo}}K_{\text{wvo}})$. This is a caveated single-galaxy preliminary endpoint, not a population validation claim.

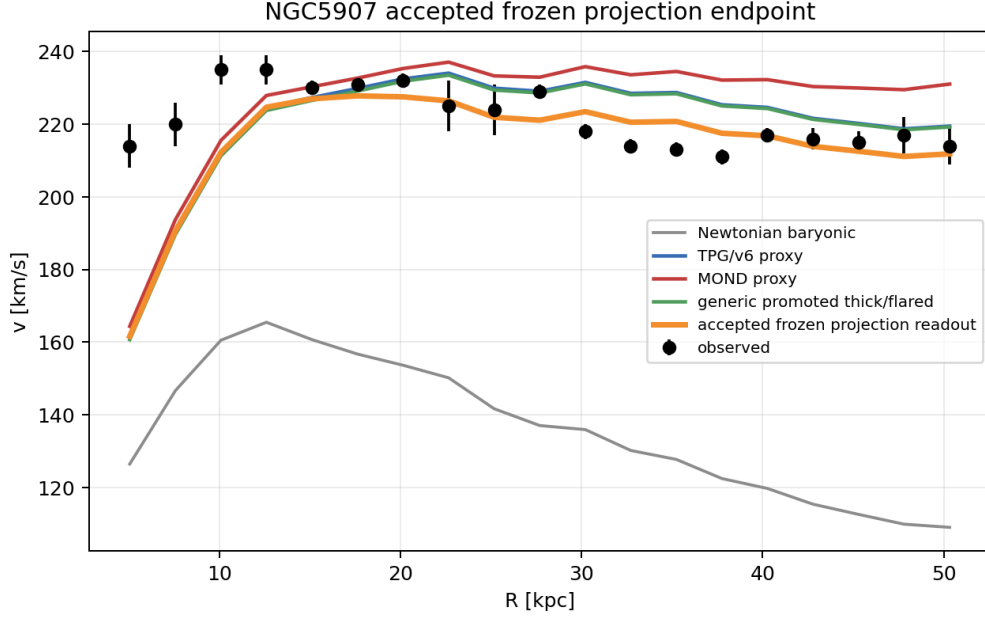


Figure 7: Projection-dominated endpoint for NGC5907. The black points are the observed rotation curve. The gray, blue, and red curves are the Newtonian, TPG/v6, and MOND proxy baselines. The purple curve is the generic promoted thick/flared proxy. The orange curve is the explicit projection-dominated Tau Core readout $v_{\text{proj}}^2 = v_{\text{TPG}}^2(1 - \gamma_{\text{proj}}K_{\text{proj}})$, with γ_{proj} and K_{proj} frozen before endpoint scoring. The result is an accepted single-galaxy preliminary control endpoint, not a population validation claim.

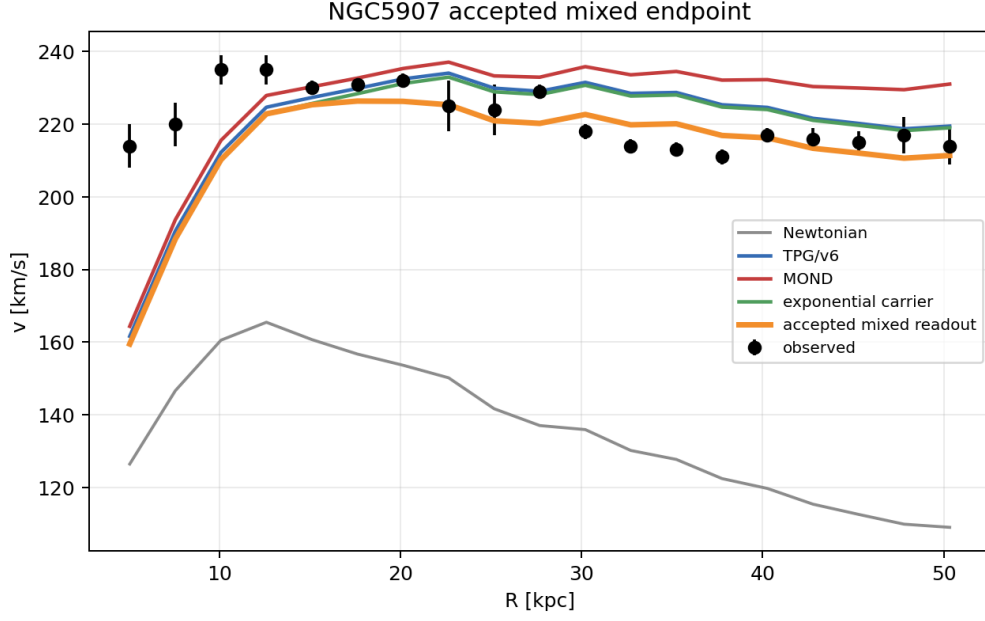


Figure 8: Accepted mixed endpoint for NGC5907. The orange curve is the source-frozen exponential-disk plus projection readout $v_{\text{mix}}^2 = v_{\text{expdisk}}^2(1 - \gamma_{\text{proj}}K_{\text{proj}})$. This row is promoted separately from the earlier projection-only endpoint: the prior projection endpoint is not used as mixed-readout evidence, and the mixed formula is read unchanged by the endpoint scoring script.

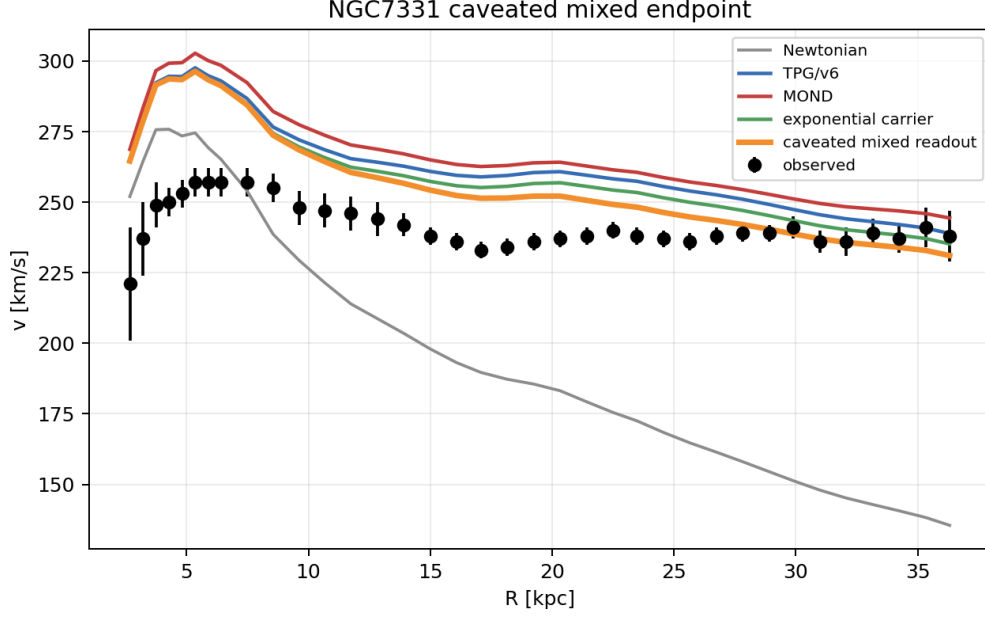


Figure 9: Caveated accepted mixed endpoint for NGC7331. The orange curve is the frozen V1 exponential-disk plus vertical/outer-warp attenuation readout. The row is endpoint-scored only after the accepted mixed gate passes, but the broad outer-window caveat remains attached. A separate V2 replay freeze manifest now records a candidate fractional-onset window; it is not applied retroactively to this score.

The package closes this preparation chain with a consolidated P0 review-pipeline status dashboard. The dashboard reports thirteen stages and the current overall decision `P0_CODEX_SOURCE_REVIEW_LABELS_READY_FULL_ENDPOINT_BLOCKED`: the P0 Codex/source-reviewed audit-label lane is populated, while full endpoint labels and endpoint scoring remain disabled.

11 Conclusion

Paper 3 is the natural launching point for this forward gate because it already contains the candidate/control framework and the required- S_τ diagnostic. The present contribution is to state the next gate in a falsifiable form:

$$\begin{aligned}
 &\text{observed morphology handle } K_{\text{obs}} \\
 &\rightarrow \text{readout proxy } K_{\text{readout}} \\
 &\rightarrow \text{formula shell } \mathcal{F}_{K_{\text{readout}}} \\
 &\rightarrow \delta g^{K_{\text{readout}}} \rightarrow \text{matched vs wrong family endpoint.}
 \end{aligned}$$

The claim-safe breakthrough target is not a universal fit claim. It is a morphology-specific prediction claim: the correct Tau Core morphology readout family, chosen before endpoint scoring, should improve residual structure more than wrong morphology families and shuffled morphology labels, while remaining competitive against Newtonian baryonic, MOND-simple, and empirical RAR baselines.

The executable package has now reached a sharper preliminary status than the original proposal: it contains four inspected readout cases, split into a three-case mixed-readout control packet,

accepted NGC5907 and caveated NGC7331 mixed endpoint promotions inside that packet, and one additional caveated NGC4088 accepted endpoint. In all four inspected rows, the matched Tau Core readout beats the best local baseline and the inspected wrong-family controls. This is not yet population validation, but it is the first compact evidence packet for the paper’s intended claim: source- or morphology-matched readout structure can matter in a way that generic baselines and wrong readout families do not automatically reproduce.

The NGC4013 mixed-readout lane illustrates the intended standard. The paper now separates three statements that should not be conflated: source evidence supports a mixed smooth-disk plus warp/vertical-overlay interpretation; the approximate mixed 4D readout formula can be frozen residual-blind as $v_{\text{mix}}^2 = v_{\text{expdisk}}^2(1 - \Gamma_{\text{wvo}}K_{\text{wvo}})$; and the already inspected numerical score is not a retroactive accepted endpoint. The new blocker gate makes that last point executable rather than rhetorical: NGC4013 is protocol-ready for future use, but accepted endpoint status requires a predeclared replay/holdout lane or a future source-selected case. This is the kind of disciplined intermediate result the program needs before population-level endpoint claims.

That is the first clean endpoint where the Tau Core bridge can become more than residual commentary.

References

- [1] A. Bosma. A 21 cm study of the galaxy ngc 7331. NED Level 5 reproduction of Chapter 4.7, 1978. Accessed for residual-blind NGC7331 H I warp source fields.
- [2] S. Comerón, B. G. Elmegreen, J. H. Knapen, K. Sheth, J. L. Hinz, M. W. Regan, A. Gil de Paz, J.-C. Muñoz-Mateos, K. Menéndez-Delmestre, M. Seibert, T. Kim, T. Mizusawa, E. Laurikainen, H. Salo, D. A. Gadotti, J. Laine, S. Erroz-Ferrer, L. C. Ho, D. Zaritsky, E. Athanassoula, A. Bosma, A. Y. K. Bouquin, E. Schinnerer, S. Meidt, and B. Holwerda. The unusual vertical mass distribution of ngc 4013 seen through the spitzer survey of stellar structure in galaxies (s4g). *The Astrophysical Journal Letters*, 738(2):L17, 2011.
- [3] W. J. G. de Blok et al. A 5 deg x 5 deg deep h i survey of the m81 group – ii. h i distribution and kinematics of ic 2574 and hijass j1021+68. *Monthly Notices of the Royal Astronomical Society*, 493(2):2618–2635, 2020.
- [4] Jozsef Olcsak. Sparc projection-sensitive residual candidate paper 3 reproducibility package, 2026.
- [5] Jozsef Olcsak. Sparc residual-disturbance paper 1 reproducibility package, 2026.
- [6] Jozsef Olcsak. Sparc residual-disturbance paper 2 reproducibility package, 2026.
- [7] Narendra N. Patra. Molecular scale height in ngc 7331. *Monthly Notices of the Royal Astronomical Society*, 480(4):4369–4376, 2018.
- [8] F. J. Sánchez-Salcedo and A. M. Hidalgo-Gómez. A new look at the holes of ic 2574. *Revista Mexicana de Astronomía y Astrofísica*, 38:39–51, 2002.
- [9] Toshiyuki Sasaki. Surface photometry of the warping edge-on galaxy ngc 5907. *Publications of the Astronomical Society of Japan*, 39(6):849–878, 1987.
- [10] Z. Shang et al. Ring structure and warp of ngc 5907: Interaction with dwarf galaxies. *The Astrophysical Journal Letters*, 504:L23–L26, 1998.

- [11] R. A. Swaters, R. Sancisi, T. S. van Albada, and J. M. van der Hulst. The rotation curves shapes of late-type dwarf galaxies. *Astronomy and Astrophysics*, 493:871–892, 2009.
- [12] J. van Eymeren, E. Jütte, C. J. Jog, Y. Stein, and R.-J. Dettmar. Lopsidedness in whisp galaxies. i. rotation curves and kinematic lopsidedness. *Astronomy and Astrophysics*, 530:A29, 2011.
- [13] Marc A. W. Verheijen and Renzo Sancisi. The ursa major cluster of galaxies. iv. h i synthesis observations. *Astronomy and Astrophysics*, 370:765–867, 2001.
- [14] Fabian Walter and Elias Brinks. Holes and shells in the interstellar medium of the nearby dwarf galaxy ic 2574. *The Astronomical Journal*, 118:273–301, 1999.
- [15] Theresa Wiegert et al. The interstellar medium and star formation in edge-on galaxies. ii. ngc 4157, ngc 4565, and ngc 5907. *The Astronomical Journal*, 150:81, 2015.
- [16] Laura K. Zschaechner and Richard J. Rand. The h i kinematics of ngc 4013: A steep and radially shallowing extra-planar rotational lag. *The Astrophysical Journal*, 808(2):153, 2015.